

DISEASES AND DISEASE CONTROL

Awareness about disease and disease control measures is essential for prevention, early detection and is key to ensuring effective treatment.

Veterinary medicine is a medical specialty concerned with the prevention, control, diagnosis, and treatment of diseases affecting the health of domestic and wild animals and the prevention of transmission of animal diseases to humans.

Animal disease is an impairment of the normal state of an animal that interrupts or modifies its vital functions. Any deviation from health is disease.

Diseases may be classified in various ways, such as:

1a) Hereditary, Congenital and Acquired Diseases:

Hereditary disease is any disease transmitted from parents to offspring eg. Hip dysplasia in dogs, Hypotrichosis in cattle

Congenital disease must be distinguished from a hereditary one. It is a disease which is acquired in utero and it is appreciable at birth. eg. Congenital heart defects, Hydrocephalus

Acquired disease is neither hereditary or congenital and most frequently develops after birth. Most of the diseases are acquired. eg. infectious diseases, deficiency diseases

1b) General and Local Diseases:

A general disease is one in which systemic disturbance receives a local change or is early associated with it .

Local disease is one in which local change is primary in some organ or part of the body and remains an essential and predominant feature throughout. eg. skin disease

1c) Structural and Functional Diseases:

Structural disease is one in which there is appreciable alteration in some organ or part of the body. eg. Structural defects of the heart

Functional disease is one in which certain symptoms are shown without any appreciable change in organs of the body which can be connected with the symptoms. eg. Epilepsy

1d) Contagious and Non-Contagious Diseases:

Infectious disease:

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Infectious diseases are diseases caused by infectious agents. The four different categories of infectious agents are bacteria, viruses, fungi, and parasites.

Contagious diseases:

These are caused by microorganisms, which spread rapidly and transfer rapidly from one animal to another by direct or indirect contact. These diseases are mainly caused by viruses. eg: - FMD, RP, Brucellosis etc.

All contagious diseases are infectious in a way but not vice versa. eg:- FMD, RP, Brucellosis etc. are infectious and contagious.

Tetanus is infectious but not contagious.

2. Classification of diseases based on occurrence:

2a) Sporadic: Occasional cases occurring at irregular intervals

2b) Enzootic: Continuous occurrence at an expected frequency over a certain period of time and in a certain geographical location (Regular cases occurring in an area)

2c) Epizootic: Refers to a sudden increase in the number of cases, above what is normally expected in the population of a particular area (Unusually high number of cases in an area)

Outbreak carries the same definition of epizootic, but is often used for a more limited geographic area.

2d) Panzootic: refers to an epizootic that has spread over several countries or continents, usually affecting a large number of animals (A global epidemic)

These terms are analogous to **endemic, epidemic and pandemic in humans**

Incidence of contagious diseases in Kerala

During winter months, sporadic outbreaks of FMD occur. So biannual vaccination against FMD is advised to boost up immunity. Use of polyvalent vaccines has decreased the number of outbreaks.

Ranikhet disease in backyard poultry has been controlled by half yearly vaccination.

Incidents of Hemorrhagic Septicemia, Black Quarter etc. are occasionally seen after the onset of monsoon but timely diagnosis and proper treatment of affected animals, preventive vaccination in and around the vicinity have reduced the outbreaks and the diseases brought under control.

Prophylactic vaccination of domesticated dogs and sterilization of female dogs (ABC) etc. to reduce the incidence of rabies in dogs are being done. But stray cases of rabies infection (Street virus) from reservoir hosts are still seen as sporadic cases.

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Outbreaks of Lumpy skin disease, Avian influenza and African swine fever have been reported recently.

Timely prophylactic measures like annual vaccination in enzootic areas, isolation of affected animals, strict control over movement of animals through the check posts, timely diagnosis and treatment of affected animals, and educating the general public about animal diseases by extension methods will go a long way in preventing infectious and contagious diseases.

Zoonotic diseases

A zoonotic disease is a disease or infection that can be transmitted naturally from vertebrate animals to humans or from humans to vertebrate animals. eg: -Rabies, Leptospirosis, Tuberculosis, other agents like prions in BSE

Health

In animals, health may be defined as the absence of disease or the normal functioning of an organism and normal behaviour based on the observation of a certain number of individuals that determine the standard and thus health.

According to World Organisation for Animal Health (WOAH founded as OIE) animal welfare means the physical and mental state of an animal in relation to the conditions in which it lives and dies. The guiding principles which inform the WOAH's work on the welfare of *terrestrial animals* include the 'Five Freedoms'. Developed in 1965, and widely recognised, the five freedoms describe society's expectations for the conditions animals should experience when under human control, namely:

Freedom from hunger, malnutrition and thirst; Freedom from fear and distress; Freedom from heat stress or physical discomfort; Freedom from pain, injury and disease; Freedom to express normal patterns of behaviour.

Signs of health in animals

A healthy animal should have:

- 1) Bright eyes and alert responding ears.
- 2) Smooth hairs of coat and a natural gloss or bloom.
- 3) Loose and movable skin over the muscle with a natural sheen.
- 4) Lick marks on the coat.
- 5) Cool and moist nose and muzzle.
- 6) Normal posture and should walk freely.

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- 7) Normally held head and not drooped.
- 8) Should move together in-group.
- 9) Should lie down and get up without difficulty.
- 10) Body temperature changes

Temperature: Body temperature is the measure of body heat of the animal and it remains constant in healthy resting animals. May vary within certain limit in animals due to age, sex, breed, atmospheric temperature, season, time of the day, exercise, feeding, watering etc. Fear, exercise, excitement, pain, etc. will raise the temperature. In advance pregnancy temperature is elevated.

Hyperthermia is simple elevation of the temperature past the critical point, as in heat stroke

Fever or pyrexia is the state where hyperthermia is combined with toxemia, as in most infectious diseases

11) Pulse & respiration : Respiration should be easy, noiseless, partly thoracic and partly abdominal. Rate, rhythm, type etc are to be observed.

Pulse: It is the rhythmic dilation of arteries that results from beating of the heart. It is counted as a number per minute. The rate of pulse will be normal in normal health. Pulse may be higher during or after exercises, excitement, fever and inflammation. It may be lower in anaemia and weakness. Animals may be made to rest before taking the pulse. Animals should be handled gently as excitement will raise the pulse. In animals the pulse is taken (1) on the underside of the base of the tail by palpating the middle coccygeal artery. (2) Under the chin by palpating the sub maxillary artery. 3) In goats, pigs, dogs and calves the femoral artery high up inside the thigh halfway between hip and stifle .

Respiration

The respiration may change in fever and diseases associated with the respiratory system. It is counted, by looking at the nostril and keeping the hands in front of the nostril or by looking at the hollow of the flank (a raise and downfall of the flank make one respiration)

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NORMAL VALUES OF TEMPERATURE, PULSE AND RESPIRATION

Species	Temperature	Pulse Per Minute	Respirations Per Minute
Cattle	100.5—102.5 (101.5 °F)	60—70	15—25
Calves	102.0—103.0 (102.0 °F)	70—90	25—30
Goats	101.5—105.0 (103.0 °F)	70—80	12—20
Pig	101.5—103.5 (102.5 °F)	60—80	10—20
Dog	100.5—103.5 (102.0 °F)	70—120	10—30
Cat	100 °F – 102.5 °F	100—130	20—30

12) Consistency in production and work :

Cows in milk should show a consistency in milk yield and working animals, a consistency in their work.

13) Dung and urine : colour, consistency etc

Urine should be clear or straw coloured. Quantity of urine in 24 Hrs: cattle:- 4-5 litres

goat: - 0.5-1.5 litres; dog:- 0.25-0.5 litres

Bowel movements should be normal and dung should not be too soft or hard. Dung output in 24 Hrs:- cattle:-25-45 kg; pig:-1.0-2.5 kg; goat:-0.5-1 kg

14) Appetite should be normal and should eat and drink well. Should have a tendency to nibble at, if green grass is offered. Should finish the full ration supplied.

15) Rumination and rumen motility

Rumination should be normal (6-8 Hrs/day). Rumen motility in cattle:- 1-2 per minute

16) Consistency in milk production

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IN GOATS: Goats should be alert with clear bold eye and head and neck held upright with good appetite and regular rumination.

IN PIGS: Interest in food is an important sign of health in pigs. Young pigs should be active and should move freely when roused. Snouts should be moist and slimy. Tail usually coiled except when lying down.(stretched out on the floor).

IN BIRDS: Their eyes should be bright, comb and wattle smooth, glossy and bright red, should be active and head held high.

General measures for dealing with outbreaks of contagious and infectious diseases

They are the following

1. Isolation or segregation
2. Quarantine
3. Disinfection
4. Destruction of affected animals
5. Disposal of carcass
6. Notification
7. Immunization

1. ISOLATION

Animals, which are known to be or suspected to be affected with a contagious disease, must be isolated from the healthy ones. A separate building is desirable, but if it is not available, the isolated animals may be housed as far away as possible from healthy ones. Feeding and management of these animals should be done by separate Attendants. Utensils and other appliances used for these animals should not be used for healthy ones and should be disinfected after use. The animal home should be disinfected. The period of isolation must be beyond the recovery of the animals and till all the possibility of the transmitting infection has passed off.

2. QUARANTINE

Is the separation of apparently healthy animals from those animals, which are healthy and not exposed to risk of infection. The object of this is to ascertain whether these animals harbor infection or not .The period of quarantine is decided considering the incubation period of different diseases. For example: Foot and Mouth disease has a short incubation period and the quarantine needed is 14 days. Quarantine is generally done when new animals are added to the

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already existing herd or the animals are transported to other places or countries.

3. DISINFECTION:

Means destruction of pathogenic microorganisms and their spores thereby the animal house and articles are made free from infection. Since the microorganisms are hidden in the dust, dirt etc. the surroundings are to be properly cleaned before disinfection.

Disinfectant: A disinfect is an agent that will kill the microorganisms and their spores without damaging the substances, with which it comes into contact. There are physical and chemical disinfectants.

Antiseptic: Is an agent that will control or kill the microorganisms. The former is used in infected places and articles, while the latter is used on animal bodies.

Physical disinfectants:

Fresh air, wind, sunlight, and heat have disinfecting properties.

Fire: Infected carcass, bedding, excreta etc. can be completely burned.

Flame: (Blow lamps) is a disinfectant for small areas and articles used in poultry houses.

Hot air oven: is used for disinfecting equipments and instruments (Temp 150-170 °F) for one hour)

Steam: is an efficient disinfectant disinfecting and sterilizing agent for equipment (autoclave at a temperature of 120 °C at 15 pounds pressure for 15 minutes).

Boiling water : Also destroys microorganisms (100 °C for 30 minutes)

Chemical disinfectants:

There are many chemicals, which have germicidal (killing of microorganisms) action and hence used as antiseptic and disinfectants.

1. Potassium Permanganate (KMnO_4): used as a mouthwash, vaginal douche and for wound cleaning. 1 in 1000 as antiseptic for genital passages and mouth, 1% as a disinfectant
2. Acid Boric 4% solution as antiseptic for mouth and 1-2% for eye irrigation.
3. Acriflavine 1 in 1,000 as an antiseptic for wounds; 1 in 10,000 to 1 in 5,000 for instruments and 1 in 10,000 for intra mammary use
4. Iodine Solution 1 in 1,000 to 1 in 500 for intrauterine use as antiseptic.
5. Argylol 2 to 5% for eye irrigation
6. Formaldehyde 5% solution is disinfectant. 1 in 400 solution is used for footbath.

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Formalin is mixed with potassium permanganate to liberate formaldehyde gas and used for disinfecting incubators (fumigation.)

7. Lysol 0.5 to 1.0 % is antiseptic and 2 - 3% is disinfectant.
8. Dettol 0.5 to 1.0% is antiseptic
9. Bleaching powder is a disinfectant and deodorant used in buildings as powder or as a solution. Normally, the fresh bleaching powder contains 25-30% available Chlorine. Usually 0.5 % (5000 ppm) chlorine solution is used for disinfecting excreta and spills of blood or body fluids; 0.1 % (1000 ppm) chlorine solution for surfaces, medical equipment and beddings.
10. Lime used for disinfecting carcasses, floors and walls of buildings and grounds. Usually 2-5 % solution of Calcium hydroxide (Ca(OH)_2), commonly called slaked lime, is used. In the powder form use 4-7 kg/100 sq.ft
11. Washing Soda sodium carbonate (Na_2CO_3) used for disinfecting utensils, floors, poultry houses etc. A hot 4% solution is used in Foot and mouth disease as a disinfectant.

Disinfection in outbreaks

Prevent the entry of unauthorized persons and animals to the animal homes, where the outbreak has occurred. Remove the excreta of animals and dispose of them properly. Clean the walls, floors, and other utensils with hot 4% washing soda and apply a disinfectant and allow it to act for 24 hours. Keep the doors and windows open so that the home will dry up. If the outbreak has occurred in a place where concrete flooring is not done, the top soil for about 4 inches is removed and disposed of. Infected pastures should not be allowed to be grazed and lime may be applied as disinfectant.

4. DESTRUCTION OF AFFECTED ANIMALS

Animals, which are severely affected and irrecoverable, should be destroyed (euthanasia) and properly disposed of.

5. DISPOSAL OF CARCASS

Proper disposal of carcasses of animals affected with infectious and contagious diseases is important to prevent the spread of infection. Disposal of carcasses may be done by incineration or burial. For the disposal of dead birds rendering or composting can also be done.

Burial

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The site chosen for the burial of a carcass should be far away from human and animal homes, streams, ponds, rivers, etc. The carcass should be buried at least 6 ft below the ground level and covered with disinfectants like lime. In some cases the carcass is cut to make the skin useless and thereby avoiding it from digging up and being used for hide. But this should not be done in the case of most infectious diseases, especially Anthrax. The natural openings of the carcass should be plugged with the cotton dipped in phenol to avoid contamination.

For disposing dead birds, deep trenches or scientifically made pits favouring microbial decomposition is cost effective, convenient and biologically safe.

6. NOTIFICATION

It is an intimation to the officers of the Animal Husbandry Department , Local Bodies and the local people for effecting control and observing proper care of the animals.

7. IMMUNISATION

Immunisation is the process by which animals are made immune or resistant to diseases. The immunity or resistance can be produced artificially by the administration of vaccine or antisera. It is the most effective method for controlling contagious and infectious diseases. It should not be considered as a substitute for other methods of control such as isolation, disinfection etc. Other methods of control are to be adopted along with immunisation.

Immunity : Immunity can be defined as all the physiological mechanisms that enable an individual's body to recognise materials as foreign and to neutralise, eliminate or metabolise them without injury to its own tissue. It can be described as the power of the body to resist infection with disease-producing organisms like bacteria, viruses or the toxic products produced by them.

Two types of immunity exist.

1) **Active immunity:** is created by its own immune system when exposed to a potential disease causing agent. Active immunity can be acquired through natural immunity or vaccine-induced immunity.

- Natural immunity is acquired from exposure to the disease organism through infection with the actual disease.
- Vaccine-induced immunity is acquired through the introduction of a killed/weakened form of the disease organism or toxoid through vaccination.

Either way, if an immunized system comes into contact with the same infectious agent in future,

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the immune system will recognize it and immediately produce antibodies needed to fight it. Active immunity is long-lasting, and sometimes life-long.

2) **Passive immunity:** occurs when the body is protected from a pathogen by immunity gained from an external source.

- A newborn acquires passive immunity from its mother through the placenta and colostrum. Certain antibodies formed in a pregnant animal will be transferred to the foetus through the placental circulation. The antibodies from a freshly calved animal will be passed to the newborn through the colostrum.
- Passive immunity can be provided through the inoculation of antiserum, when immediate protection from a specific disease is needed.

The major advantage of passive immunity is immediate protection whereas active immunity takes time (usually several weeks) to develop. However, passive immunity lasts only for a few weeks or months whereas active immunity is long-lasting.

Antigen: An antigen is a foreign substance which stimulates the immune system to produce antibodies against it.

Antibody: An antibody is a protein produced by the body's immune system in response to a specific antigen.

The antibodies are produced by the immune system in response to vaccination or natural infections.

Vaccine: Vaccine is a biological preparation containing antigens (killed/weakened microbes, purified antigens or toxins) that induces the production of antibodies against specific diseases by stimulating the body's immune system.

Antiserum is the blood serum that contains antibodies against an infectious agent (bacteria, virus etc.) or toxic substance (tetanus toxin, snake venom etc.) used to prevent or treat infection or poisoning.

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Antiserum can be produced by inoculating a sublethal dose of antigen (disease causing microbe/toxin). The antiserum can be used for giving immediate protection to an animal in outbreaks since it contains readymade antibodies. The establishment of immunity in antiserum inoculation is immediate but lasts for only a few days (10—14 days). It can be used in outbreaks for controlling diseases. Some antiserum, like anthrax antiserum, anti tetanus serum, anti rabies serum , anti snake venom serum etc. is used for curative purposes.

Vaccination and Inoculation

Vaccination is the process of administering a vaccine to produce immunity against that disease. Vaccines can be administered subcutaneously, intramuscularly or by other methods like intranasal, oral, wing web puncture etc. Vaccination gives prolonged or permanent immunity against diseases, but it takes about 10 to 14 days for the development of immunity.

Inoculation is the process by which antiserum is administered to an animal. The term is also used for administering infected material to the laboratory animals for diagnostic purposes or for producing antiserum.

Serum Simultaneous method of control or immunization is the simultaneous administration of antisera and vaccine. Since antisera gives only temporary protection, the vaccine is also administered along with antiserum to give permanent immunity.

Infectious agents that cause diseases in animals fall into five groups

- 1.Bacteria
- 2.Virus
- 3.Fungus
- 4.Protozoa
5. Helminths

DISEASES OF CATTLE

BACTERIAL DISEASES OF CATTLE

Anthrax (Wool Sorter's disease: splenic fever, Charbon)

Malayalam: (അടപ്പൻ)

It is a Peracute/ acute, febrile infectious disease affecting all animals except birds. It is a

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zoonotic disease. It is widespread throughout the globe. The causative agent is *Bacillus anthracis*, a rod shaped bacilli. The bacilli are non-motile, very large rod-shaped, capsulated, gram positive and spore bearing. Anthrax bacilli form a hard, protective coating called spore when they are exposed to atmospheric air. The spores thus formed remain viable in the soil for long periods. Never open a carcass for investigative purposes, if it is suspected that the death was due to Anthrax. The organisms are free to escape to the soil and be there in a sporulated form for years together, to reinfect the cattle when a favorable condition arises. The disease affects all species of animals. The mode of infection in cattle is by

1. Ingestion of contaminated feeds and water by grazing on infected pastures.
2. Through wound infection (rare)
3. Inhalation and lastly by
4. Vectors like biting flies (*Tabanus* sp.)

Incubation period is 1 to 14 days (usually 2-3 days)

Symptoms

In peracute cases sudden death without any clinical signs is usually noticed (death within 1-2 hours). In acute cases high fever at the onset (105 °F -107°F), dullness, depression, off feed, congested mucous membrane, abdominal pain, convulsion and death are the main symptoms. The carcass will be highly distended and bloated up. Tarry dark unclotted blood will be seen escaping from the natural orifices of the body.

Diagnosis

1. Symptoms
2. By laboratory examination of blood smear. Materials to be sent for lab examination are blood smear (2—3), fixed over flame.
3. Guinea pig/mice inoculation and reproducing the disease.
4. Ascoli's test with ear/muzzle piece.
5. Various laboratory tests like PCR, ELISA etc.

Treatment

Usually there will be no time for instituting treatment, as diagnosis will be mostly after death. Broad spectrum antibiotics are effective.

Control

- 1 Proper disposal of the carcass .The natural openings of the carcass should be plugged with

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cotton wool dipped in disinfectants (phenol). The carcass should be incinerated or buried in a pit of 6 ft depth and covered alternatively with lime and sand.

2 Yearly immunisation with Anthrax Spore Vaccine

3 By proper hygiene measures. The cattle sheds and the area where the animal was tied should be washed with strong disinfectants (10% formalin, 5% Lysol, 5% sodium hydroxide). Beddings, straw, Excreta and other materials should be burned. The utensils used should be cleaned and disinfected

Black Quarter (Black leg, Quarter ill, Emphysematous gangrene)

Malayalam : (കരികാൽ / കരിങ്കു)

This is an acute infection but not contagious disease of cattle, buffalo, goats and sheep. It is characterised by high fever and crepitating swellings in the muscles of thigh and quarter leading to lameness.

The etiologic agent is a rod shaped bacteria viz. *Clostridium chauvoei*.

The disease is mainly seen in 6 months to 2 year old animals in good condition

Modes of infection

a. Soil infection.

C. chauvoei is a sporulating organism and as such spores are formed both inside and outside the carcass.

b. Ingestion

Infection occurs when cattle graze on contaminated pastures.

c. Wound infection

Is also possible through contamination

Incubation period

2 to 5 days. The onset is sudden.

Symptoms

Animals sometimes die without any premonitory symptoms. Feverish animals show lameness/stiffness, depression, and anorexia. Swellings develop in the muscles of the shoulder, hip, chest, back or flank and sometimes in the neck. Affected areas show trembling and violent twitching. The swelling is small, hot and painful in the beginning. In a few hours, it becomes extensive, crepitating, less sensitive, cool and painless. The skin over the swelling may become

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dark, dry and parchment like (paper like). Finally the temperature falls below normal and the animal becomes exhausted, succumbing to death, which occurs due to toxæmia.

Diagnosis

1. From Symptoms
2. Demonstration of organisms in edema fluid.
3. Various laboratory tests

Control

1. Isolation of affected animals
2. Proper disposal of carcasses
3. Prophylactic vaccination

Treatment

Antibiotics can be tried but of little use.

Haemorrhagic septicaemia (Pasteurellosis; shipping fever)

Malayalam: കരലടപ്പൻ

It is an acute/ sub cute, infectious and contagious disease affecting cattle, buffaloes, sheep, goats and pigs, characterised by high fever and respiratory problems. The causative organism is *Pasteurella multocida*. The disease is seen as seasonal outbreaks after the onset of rain. Buffaloes are more affected. Factors, which lower the body's resistance like change in climate, malnutrition etc., predisposes to infection.

P.multocida is an organism, which is usually seen in the throat region of healthy animals. It is also assumed that such normal inhabitants of the respiratory tract become virulent when the general resistance of the animal is lowered due to external causes like transportation. Hence the name Shipping Fever.

Route of infection:

1. By ingestion
2. By inhalation.

Incubation period is 2—5 days.

Symptoms

High fever (105 °F—108°F) dullness, inappetence, labored breathing, nasal discharges, rapid respiration, abdominal pain, salivation, hot painful swelling of the throat and dewlap and in acute cases septicaemia and sudden death.

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Diagnosis

1. From symptoms
2. By blood smear examination and other laboratory tests

Control

Proper disposal of carcasses, isolation, disinfection of premises are important. Immunisation of healthy animals using H.S Broth vaccine/Oil adjuvant vaccine before onset of monsoon. In clinical cases (subacute cases) treatment with appropriate antibiotics can be done.

Tuberculosis

Malayalam: (ക്ഷയം)

A chronic infectious disease caused by *Mycobacterium tuberculosis* and characterised by the development of nodules of granulation tissue or tubercles, which undergo calcification, caseation or abscess formation. The organs so affected are mainly lymph glands. Depending on the organ affected the infection may be designated as pulmonary tuberculosis (lung) or intestinal tuberculosis etc. Almost all visceral organs are susceptible to the infection. Besides tuberculosis can affect all systems of the body simultaneously called miliary tuberculosis (1-2 mm tubercle foci). Miliary tuberculosis is a potentially fatal form of the disseminated disease due to the hematogenous spread of tubercle bacilli to the lungs, and other organs. In animals it is rare. Tuberculosis is a zoonotic Disease.

M. Tuberculosis is an acid-fast Bacillus (Ziehl-Neelsen's stain). It is so called because of the staining peculiarity. It affects cattle, swine goats and poultry. There are three main strains of M. tuberculosis viz Bovine, Avian and Human.

Mode of infection:

1. By ingestion; overcrowding and unhygienic surroundings favour the infection.
2. By inhalation
3. Fresh Milk from affected animals.

Incubation Period: is very long since it is a chronic infection.

Symptoms

Poor body condition and weight loss is the most frequent indication of Tuberculosis. This is a general symptom (Emaciation) and there are other symptoms depending on the organ affected. If lungs are affected there are respiratory symptoms like difficulty in breathing, harsh dry cough, nasal catarrh, bronchitis etc. In intestinal TB, emaciation and diarrhoea are seen.

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Though the local lesions of tuberculosis are seen in superficial Lymph glands, the generalised lesions occur in lungs, liver, intestine, pleura, and udder (Tuberculous mastitis).

Diagnosis:

1. From symptoms
2. By demonstrating acid- fast- bacilli in the discharge (phlegm) or other samples depending on the organ affected.
3. Intradermal Tuberculin test is used for screening.

Control:

Best management practices like proper hygiene, sanitation and avoiding overcrowding.

Testing and disposal of infected animals.

Great care should be taken when new stocks are introduced.

Johne's disease (Para tuberculosis)

Johne's disease is a highly fatal chronic, contagious granulomatous enteritis in cattle and other ruminants by progressive weight loss, debilitation, and eventually death.

Aetiology: *Mycobacterium paratuberculosis* an acid- fast- bacteria akin to tubercle bacilli

Mode of spread:

The disease spreads by ingestion of feed and water contaminated by the faeces of infected animals. The infection occurs mostly in the early months of life. The incubation period extends from 12 months to several years. Affected animals may not show clinical signs and continue to discharge organisms in faeces. The organisms persist in pastures for about a year.

Symptoms

The first distinctive symptom is chronic fetid diarrhoea without straining. There is no inappetence at the beginning but the animals lose condition in spite of good feeding. Milk production gradually decreases. The disease is characterized by progressive emaciation, recurrent diarrhoea which persists for months.. As the disease progresses there is severe dehydration, wasting, anaemia and death.

Diagnosis:

1. From symptoms
2. Demonstration of organism in smears after acid fast staining (rectal pinch for acid- fast bacilli using Ziehl- Neelsen stain)
3. Intradermal test using Johnin

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Control:

1. Treatment is not recommended. Slaughtering of seropositives is advisable.
2. Disinfection and hygienic measures to control contamination
3. Testing and destruction of positive animals (Johnin Test).

White Scours. (Bacillary White Diarrhoea)

It is a disease of newborn calves (1-3 wks old) characterised by scouring, weakness and prostration caused by a bacterial organism called *Escherichia coli*. Besides *E. coli* other bacterial, viral, protozoan organisms may also contribute and failure to transfer passive immunity is considered an important predisposing factor.

E. coli is a normal habitant of the gastrointestinal tract of human beings and animals. The mortality rate is high.

Mode of infection.

1. By contaminated food and water (mainly faecal contamination.)
2. Suckling calves contract the disease from contaminated udder.
3. Through umbilicus (navel infection)

Incubation period: 1—5 days

Symptoms

In acute cases the affected calves show signs of fever initially, colic, dysentery, dehydration, pale mucosa, cold extremities in later stages (fall in temperature), convulsions, coma and die within 2-6 hours. In less acute cases calves are listless, unable to suckle, and develop foetid (foul smelling) diarrhoea. Older calves with diarrhoea pass loose whitish faeces and owners think that the white colour may be due to undigested milk. Swelling of joints and pneumonia are seen in a few cases.

Diagnosis:

1. From symptoms.
2. Laboratory examination of faeces.

Control:

Calves should be kept in a hygienic environment.

Prevent contamination of food (milk), water and premises. Applying a calf muzzle after feeding is advisable.

Navel ill & Joint ill

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These are two septicaemic diseases of the newborn animals characterised by swelling of joints and umbilicus caused by several bacteria like *Streptococci*, *E.coli* and *Shigella sp.* etc.

Mode of Infection.

1. Through umbilicus after birth
2. Intrauterine infection.

Incubation period is usually 7 days.

Symptoms:

If infection stays mostly confined to the navel, the primary sign is a swollen and painful navel that does not dry up. Affected calves exhibit high temperature, reduced appetite and an abscess may develop at the umbilicus.

If the infection spreads from the navel/ navel ill is not treated, septic arthritis, or joint-ill develop due to hematogenous transmission of bacteria to the synovial structures (commonest site is stifle and hock joint) of young animals. Diagnosis can often be made based on clinical signs of a hot, painful, stiff swollen joint. Further signs will develop as bacteria spread via the bloodstream and settle in other parts of the body.

Control

1. Proper sanitation and hygiene and ensure that the cow calves in a clean environment.
2. The umbilical cord may be severed aseptically and the site should be painted with Tr. Iodine.
3. Proper treatment of affected cases

Foot rot

Malayalam: കളമ്പുചീയൽ

This is an infectious disease of animals characterised by the inflammation, necrosis and ulceration of the inter digital space, coronary bands and posterior limbs resulting in lameness.

Aetiology:

The main foot rot-causing bacteria in cattle is *Fusobacterium necrophorum*, a ubiquitous bacteria found in the environment.

The disease is seen in cattle when they are kept in wet and muddy areas. The infection is taken up through abrasions and small wounds present in the foot of cattle.

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Symptoms:

Lameness of one or more limbs and walking on the knees are the main symptoms. The temperature rise is negligible. The affected area viz. The pastern is swollen and painful. The skin above the inter digital space is fissured, swollen and necrosed with a foul smelling discharge.

The disease can be prevented by observing strict hygiene and sanitation. Wounds around the feet should be attended immediately. This disease responds to treatment

Brucellosis (Bovine contagious Abortion, Bang's disease)

It's an acute or chronic contagious disease of cows, buffaloes, sheep and goats. Abortion is the most obvious manifestation. Infections may also cause stillborn or weak calves, retained placentas, infertility and reduced milk yield. The causative agent is a *Brucella abortus*. Brucellosis is a zoonotic disease.

It is a disease of major economic importance in cattle as it causes abortion and is of a contagious nature. The disease is communicable to human beings, through milk, uterine discharges etc. In human beings it is called undulant fever/ malta fever. Several sero types of the organism are seen and almost all of them causes abortions in animals. Viz. *B.melitensis* (sheep and goat), *B.suis*(swine), *B.canis*(dogs)

Mode of infection

1. Natural transmission occurs by ingestion of organisms, which are present in large numbers in aborted fetuses, fetal membranes, and uterine discharges. Cattle may ingest contaminated feed and water or may lick contaminated genitals of other animals.
2. By consuming milk from an infected and asymptomatic animal
3. *Brucella* may enter the body through mucous membranes, conjunctiva and wounds, in humans and animals.
4. Venereal transmission by infected bulls to susceptible cows appears to be rare.
5. Transmission may occur by artificial insemination when *Brucella*-contaminated semen is deposited in the uterus .

Incubation period varies from 2 weeks to several months.

Symptoms

The organisms mainly affect the genital organs like uterus, testis, accessory sex glands, lymph

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glands and joints. Infection spreads rapidly and causes many abortions in unvaccinated cattle. The abortion is characteristic in that it occurs after 6 months (6 to 8 months) of pregnancy. The placenta in such cases will always be retained and require manual removal. The placenta presents a leathery appearance. Bacteria may be found in the uterus during pregnancy and during uterine involution. Shedding from the vagina largely disappears after parturition with the cessation of lochial discharges. Organisms are shed in milk for a variable length of time (in most cattle for life). Some infected cows that previously aborted shed brucella from the uterus at subsequent normal parturitions. The organisms after abortion, vanishes from the uterine tissue and through circulation gets lodged in the joints causing synovitis. Mild endometritis is also seen in such animals. The cow may become temporarily infertile. If these animals are bred again, they conceive and the vicious circle is repeated again. The organisms escape from the joints in late pregnancy and infect the pregnant uterus causing abortion. Hence, treatment is not advisable and affected animals become carriers of the infection.

The organism causes Orchitis in bulls and gets transmitted through semen. This organism also causes inflammation of joints and hygroma especially in males.

Diagnosis

1. Recurrent abortions at 6 to 8 months is the typical symptom.
2. Laboratory Diagnosis by various tests (ABR/Milk-ring test, Individual blood tests).
3. Lab examination of uterine discharge, placenta and stomach contents of the aborted foetus (examination for the presence of bacteria.)

Control:

1. Efforts are directed at detection and prevention, because no practical treatment is available.
Eventual eradication depends on periodic blood testing and eliminating carriers. Herds must be tested at regular intervals until two or three successive tests are negative.
2. Stamping out of affected animals and proper disposal of carcasses
3. Immunisation using Strain 19 (calf hood vaccination) or RB - 51, Brucella abortus vaccine.

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Leptospirosis (Mud Fever/Weils disease)

Malayalam: എലിപ്പനി

Leptospirosis affects cattle, sheep, goats, pigs, horses, and dogs but is rare in cats and seen worldwide. Large outbreaks have occurred after floods.

Aetiology: Caused by spiral shaped bacteria *Leptospira* sp. (Spirochaetes)

Mode of infection:

Leptospirosis is an infection caused by the spirochete *Leptospira*. It spreads through the urine of infected animals which gets into water or soil and can survive there for weeks to months. Animals get infection through direct contact with infected urine or tissues, contaminated water and soil, abrasions in the skin and mucous membranes (eyes, nose or mouth), drinking contaminated water and rarely by inhalation (aerosol). It's a zoonotic disease

Humans can become infected with leptospirosis in ways similar to animals (direct contact, ingestion, through contaminated water or soil).

Symptoms

Symptoms are biphasic. Both phases involve acute febrile episodes; the 2nd phase sometimes includes hepatic, pulmonary, renal, and meningeal involvement. In acute disease, cattle, sheep, goats and swine exhibit high fever, petechiae of mucosa, severe jaundice, dark coloured urine, albuminuria, dyspnoea and death. In chronic form mastitis and abortions are seen.

Diagnosis

Laboratory tests (blood culture, serologic testing, PCR etc.)

Prevention and control:

Avoid contact with contaminated water, soil, contaminated environments or infected wildlife, especially rodents.

Do not allow animals to drink from or enter contaminated water bodies.

Dogs should be regularly vaccinated against *Leptospira*.

VIRAL DISEASES

Foot and mouth disease (Apthous fever) -

Malayalam: കളമ്പുരോഗം

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It is an acute highly contagious disease affecting all cloven-footed animals caused by a virus and characterised by high fever, formation of vesicles in the mouth and feet and salivation. FMD virus has several strains like O, A, C, Asia 1, SAT 1, SAT 2, SAT 3 etc. It is the most contagious of all diseases and cattle, swine, sheep and goats are mainly affected. Local (desi breeds) animals are not seriously affected but cross breeds and exotic breeds are severely affected. Mortality rate is higher in calves.

Mode of infection:

- (1) direct contact with affected animals and fomites
- (2) through contaminated food and water
- (3) through insects and animal attendants.
- (4) through aerosol

Incubation period: 1-21 days, usually 3-7 days.

Symptoms: First noticeable symptom is high fever (104°F- 108 °F), which lasts for 1-2 days. This is accompanied by salivation, smacking of lips, reddening of oral mucosa, anorexia, depression and fall in milk yield. Vesicles are formed on the dorsum of the tongue, the mucosa of lips, cheek, gum, dental pad, hard palate, around the muzzle, coronary band, interdigital cleft and udder. The vesicles rupture leaving ulcers and the salivation is increased. The saliva flows in long hanging strings (Stringy salivation). The vesicles on the inter digital space rupture forming ulcers. There is severe lameness. Rarely the vesicles appear on teats and get ruptured during milking. In swine (amplifier host), sheep and goats the disease is mild.

Diagnosis:

- 1) from symptoms
- (2) Various laboratory tests(vesicular fluid, scraping from lesions)

Control:

1. Common measures like isolation, disinfection, disposal of infectious materials and avoiding overcrowding are to be done.
2. Immunisation of animals should be done using a polyvalent oil adjuvant vaccine (Strain O, A, Asia 1)
3. Treatment of affected cases. The vesicles and ulcers of the mouth, foot and teats are to be cleaned with antiseptics (1 in 5000 potassium permanganate and 2% alum) and dressed with protective dressings [Boroglycerine for mouth and teat and a mixture of tar, copper sulphate and

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neem oil for foot]. Apply fly repellants and antiseptic dressings.

4. As a virucidal agent, 4% sodium carbonate solution can be used to disinfect cattle shed, utensils etc.

Biannual preventive vaccinations help to prevent outbreaks. Since the virus is of different serotypes, a polyvalent vaccine is advisable. In Kerala, it has been found that the yearly outbreaks have come down, after the government has started vaccinations under NADCP.

Rinderpest (Cattle Plague)

Malayalam: കാലിവസന്ത

It is an acute highly contagious disease of ruminants caused by a virus and characterised by inflammation of the alimentary tract. It is a highly fatal disease and cattle and buffaloes are usually affected.

Mode Of Infection:

- 1) contact with affected animals
- (2) through contaminated food and water
- (3) through vectors.

Incubation period: 3-9 days.

Symptoms: High fever (105-107°F) which stands for a week, frothy salivation, lacrimation, mucopurulent oculonasal discharge, staring coat and suppression of rumination are shown by the animals. The oral mucosa is reddened with grayish white spot/ulcers (appearing as if sprinkled with bran) on the lower lip, gums, cheek and tongue with profuse salivation. Following this a foul smelling shooting diarrhoea (fishy odor) appears. The temperature falls below normal and the animal becomes prostrated and dies off by 7-10 days.

Diagnosis

- 1.Symptoms and postmortem findings
- 2.Various laboratory tests

Control:

Control the movement of cattle
compulsory vaccination

Common measures of control of a contagious disease are to be adopted. Nursing and proper diet are also essential.

The mouth lesions should be washed with antiseptics and demulcent dressings should be applied.

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The disease has been declared eradicated from our country.

Ephemeral Fever (Three day sickness)

Malayalam: മുടന്തൻ പനി

Bovine ephemeral fever is an insect-transmitted, non contagious, viral disease of cattle and water buffalo. During epidemics the onset is rapid and many animals are affected within days. Morbidity may be as high as 80% but overall mortality is usually 1%–2%.

Clinical findings: Sudden onset of high fever (104°–107° F), muscular shivering, inappetence, lacrimation, serous nasal discharge, drooling, stiffness of joints and shifting lameness, and a sudden decrease in milk yield are the main symptoms. Bulls, heavy cattle, and high yielders are most severely affected, but spontaneous recovery usually occurs within a few days.

Diagnosis:

1. from symptoms
2. Various laboratory tests

Treatment and control: Symptomatic treatment is advocated. Advise complete rest for ailing animals and recovering animals should not be stressed because relapse is likely. Application of liniments, hot fomentation, antipyretics/anti inflammatory drugs given orally are effective. Drenching should be avoided.

Lumpy skin disease

Lumpy skin disease is an infectious, eruptive, occasionally fatal disease of cattle caused by a pox virus, Lumpy Skin Disease Virus (LSDV) characterized by nodules on the skin and other parts of the body.

The incubation period is 4–14 days.

Symptoms: Infected cattle develop fever, lacrimation, nasal discharge, and hypersalivation, followed by the characteristic eruptions on the skin and other parts of the body. The nodules are well circumscribed, round, slightly raised, firm, and painful and involve the entire cutis and the mucosa of the GI, respiratory, and genital tracts. Secondary infection sometimes occurs and causes extensive suppuration and sloughing. In time, the nodules either regress, or necrosis of the skin results in hard, raised areas (“sit-fasts”) clearly separated from the surrounding skin. These areas slough to leave ulcers, which heal and scar.

Control: Treatment of affected animals and Vaccination of susceptible animals

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IBR/IPV

Infectious Bovine Rhinotracheitis (IBR)/Infectious pustular vulvovaginitis is an acute, contagious respiratory disease of cattle caused by a virus, commonly affecting the respiratory tract and the reproductive system. It is highly contagious, resulting in rapid spread of respiratory disease among cattle in close confinement. It cause abortions, genital infections and vulvovaginitis in cows; balanoposthitis in bulls and sexually transmissible.

Cowpox (ഗോവപുഷ്പ) & Buffaloe pox

The disease is acute and contagious, caused by a virus and characterised by eruptions on the skin and low grade fever.

Mode of infection: is by direct contact.

In cows the lesions are noticed on the udder and teats and in bulls on the scrotum. In buffaloes, the lesions may be seen around the ears, ear flap, eyes, neck region and sometimes throughout the body.

The eruptions pass through four stages, namely; papules, vesicles, pustules and scabs. The scab falls off leaving a red patch, which then heals off. The incubation period is 8 - 12 days. The cow pox is communicable to milkers and they develop skin lesions on hands.

Treatment: The skin lesions should be dressed with ointments (Boric acid ointment, Zinc oxide ointment) or with Tincture Iodine in glycerin.

Control: By isolation, hygiene and sanitation and disinfection.

Rabies (റേബിസ് ബാധ)

Rabies is a fatal viral encephalomyelitis transmitted by the saliva of infected mammals and bats. It affects **all warm blooded animals** including man, dogs and domestic animals, characterised by disturbances of the central nervous system, paralysis and death.

Mode of infection: By the bite of rabid animals and contamination of wounds by saliva of infected animals and rarely through mucous membranes. Wild canids like foxes, jackals etc. are the principal reservoir hosts. The neurotropic virus travels from the site of entry via peripheral nerves to the spinal cord (or to the brain stem when the face is bitten), then to the brain. It then spreads from the central nervous system (CNS) via peripheral nerves to other parts of the body. Virus multiplies in the salivary gland and the salivary glands are an important site of viral replication and portals of exit for the rabies virus into the saliva.

Incubation period: varies from a few days to several months. The incubation period depends

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to some extent on the site of the bite and the quantity of virus deposited.

Symptoms:

Clinical signs of rabies are rarely definitive. Rabid animals of all species usually exhibit typical signs of CNS disturbance, with minor variations among species. The most reliable signs regardless of species, are acute behavioral changes and unexplained progressive paralysis. Behavioral changes may include sudden anorexia, signs of apprehension or nervousness, irritability, and hyperexcitability. The animal may seek solitude, ataxia, altered phonation, and changes in temperament. Uncharacteristic aggressiveness may develop, a normally docile animal may suddenly become vicious. Commonly, rabid wild animals may lose their fear of people, and normally nocturnal species may be seen wandering about during the daytime.

Two forms of the disease are seen in animals, **furious form and dumb or paralytic form.**

Furious Form

The furious form of rabies refers to animals in which aggression is pronounced. This is the classic “mad-dog syndrome,” although it may be seen in all species. There is rarely evidence of paralysis during this stage. The animal becomes irritable and, with the slightest provocation, may viciously and aggressively use its teeth, claws, horns, or hooves. The posture and expression is one of alertness and anxiety, with pupils dilated. Noise may invite attack. Such animals lose caution and fear of people and other animals.

Carnivores with the furious form of rabies frequently roam extensively, attacking other animals, including people, and any moving object. They commonly swallow foreign objects, eg. feces, straw, sticks, and stones. Rabid dogs may chew the wire and frame of their cages, breaking their teeth, and will follow a hand moved in front of the cage, attempting to bite. Young pups can seek human companionship and are overly playful, but bite even when petted, usually becoming vicious in a few hours. Rabid domestic cats can attack suddenly, biting and scratching viciously. As the disease progresses, muscular incoordination and seizures are common. Excitement followed by paralysis and death in 3-6 days following the onset of clinical signs.

Dumb/Paralytic Form

The paralytic form of rabies is manifested by ataxia and paralysis of the throat muscles, often

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with profuse salivation and the inability to swallow. Dropping of the lower jaw is common in dogs. Owners frequently examine the mouths of dogs and livestock searching for a foreign body or administer medication with their bare hands, thereby exposing themselves to rabies. These animals may not be vicious and rarely attempt to bite. The paralysis progresses rapidly to all parts of the body, and death may occur in 5-8 days.

Species Variations

Cattle with furious rabies can be dangerous, attacking people and other animals. Lactation ceases abruptly in dairy cattle. The usual placid expression is replaced by one of alertness. The eyes and ears follow sounds and movement, change in vocalization, abnormal bellowing which may continue intermittently until shortly before death, spasms of the muscles, salivation, paralysis and death are noticed.

Hydrophobia is a clinical sign characteristic of human rabies. Attempting to drink causes painful spasms of the laryngeal and pharyngeal muscles. Hydrophobia is not prominent in animals.

Diagnosis: (1) from symptoms (2) examination of the brain for Negri bodies in the laboratory; Antigen detection using fluorescent antibody test (FAT) (3) Lateral flow tests are available; not used for confirmation (4) PCR test

Control:

Notification of suspected cases. Suspected animals should be secured and carefully watched.

Control the stray dog population by Animal Birth Control Programme (ABC-AR), along with other control strategies like waste management etc.

Prophylactic vaccination and licensing should be done for all domesticated dogs.

Post exposure vaccination for animals bitten by suspected dogs.

Prophylactic vaccination in dogs and cats

First dose - 3 months of age

First booster - one month after first dose

Then annual vaccination

Post exposure vaccination (all species)

Immediately after the suspected bite, post bite vaccination should be done on days 0, 3, 7, 14, 28 and 90 (optional).

FUNGAL DISEASES OF CATTLE

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Dermatophytosis

It is the most common fungal disease affecting all warm blooded animals.

Aetiology: Trichophyton sp.; Microsporum sp.

Clinical signs: Lesions range from slight erythema to highly inflammatory folliculitis, scaling patches of hair loss with gray-white crust formation, but some become thickly crusted with suppuration and extensive areas of alopecia.

Treatment and control: Involves improvement of husbandry practices because overcrowding increases disease prevalence. Remove crusts with a brush (discard the brush) and burn the infective material. Topical therapy with antifungals is the treatment of choice.

MASTITIS IN CATTLE

Mastitis is inflammation of the udder/mammary gland with physical and chemical changes of milk and pathological change of the udder tissue. Mastitis is one of the most economically distressing diseases of dairy cattle throughout the world. Although stress and physical injuries may cause inflammation of the gland, infection by invading bacteria or other microorganisms (fungi, yeasts and possibly viruses) is the primary cause of mastitis. Infections begin when microorganisms penetrate the teat canal and multiply in the mammary gland. Hematogenous spread is also possible.

Mastitis can be clinical or subclinical.

Clinical mastitis can be divided into three types:

1. **Per-acute mastitis:** It is characterized by gross inflammation, reduction in milk yield, changes in milk composition and systemic signs like fever, off feed, depression, shivering etc.
2. **Acute mastitis:** Inflammation of the gland will be severe, but with lesser systemic signs like fever and mild depression.
3. **Sub-acute mastitis:** Inflammation of the gland is minimal and no visible systemic signs.

Chronic mastitis: Inflammation is mild, gland not swollen, pain and heat absent, induration and clots in watery fore milk. In chronic mastitis inflammatory process exists for months and may continue from lactation to lactation. It exists as subclinical but may exhibit periodical flare-ups subacute or acute form, which last for a short period of time

Subclinical mastitis: The subclinical mastitis is characterized by reduction in milk yield without any visible signs either in the udder or in the milk. The incidence of subclinical mastitis is 15 to

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40 times more than the clinical form. It adversely affects milk quality and production because of longer duration, difficult to detect and constitutes a reservoir of microorganisms that lead to chronic mastitis condition and infection of other animals within the herd.

Gangrenous mastitis: Skin of the quarter and teat is semi - cold and bluish and may slough off

Diagnosis

1. California Mastitis Test (CMT) for diagnosing subclinical mastitis.
2. From symptoms

California Mastitis Test

Mastitis milk contains a large number of somatic cells. CMT reagents contain detergents and a pH indicator. Mastitic milk tends to gel when tested by CMT. The detergents in the reagent react with DNA in the somatic cells resulting in gel formation. The degree of gelling indicates the presence and severity of mastitis. pH change of the mastitis milk is reflected by the indicator in the reagent. The change in colour indicates the pH variation of the milk and therefore, the level of inflammation.

Procedure: After discarding the first stream of milk, draw the milk into the shallow cups on the paddle, keeping the quarters separate. Always assume the same position when holding the paddle under the udder to keep track of the quarters when interpreting.

Draw foremilk from each quarter (around 5 ml) into separate cups of a four-cup plastic paddle. Add an equal amount of CMT reagent. Proportion of solution to milk should be at least one to one (1:1). Mix the reagent and milk by gently rotating the paddle in a horizontal plane, swirling the mixture for 10-30 seconds. Milk from a normal quarter remains liquid and flows freely. In positive cases, there will be colour change and gel formation and it can be graded during the rotary motion. In a moderate reaction, the gel that forms is fragile and breaks into small masses or clumps. Milk samples that react strongly form a gelatinous mass that clings together as the paddle is rotated. CMT is a measure of inflammation within a quarter and can be scored on the basis of colour change and degree of gel formation(Negative, Trace,1+,2+,3+ etc).

Control & Management

- Proper dry cow management; dry cow therapy, internal and external teat sealants during the dry period etc.
- Housing; Provide adequate space, ventilation, bedding, and lighting to ensure cleanliness

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and comfort at all times. Overcrowding should be avoided.

- Clean and dry environment (avoid manure buildup)
- Fly control, elimination of fly breeding sites
- Maintain proper hygiene of the udder; clip the hairs on the udder, clean udder, dry teat and teat ends etc
- Hygiene of the milkers; hand washing, Prevent transfer of pathogens from cow to cow during milking
- Milking machine maintenance
- Proper milking procedure and cows should be milked completely, Standard milking techniques should be used like; apply milking units properly and make adjustments to prevent the admittance of air into the teat cup liners and prevent liner slip. Prevent injury to the teats and udder during milking
- Teat dipping
- Early detection of new infections (clinical and subclinical) and treatment
- Culling cows with chronic mastitis
- Do not allow the animal to lie down immediately after milking
- Provide adequate nutrition to preclude increased susceptibility to mastitis . Correct minerals and vitamin deficiencies(copper, zinc, Selenium, Vit.A, E, D)

Success in the management of mastitis depends on the nature of the aetiological agent involved, the severity of the disease and the extent of fibrosis. Complete recovery with freedom from bacterial infection can be obtained in cases of recent infection and in those where fibrosis has taken place only to a small extent. Treatments usually done with antibiotics, anti-inflammatory drugs and antipyretics. Indiscriminate use of antibiotics in mastitis treatment should be avoided since it causes human health hazards through antimicrobial resistance(AMR) .

DISEASES OF SWINE

BACTERIAL DISEASES OF SWINE

Swine erysipelas (Diamond skin disease)

This is an infectious disease affecting pigs and is caused by a bacteria viz. *Erysipelothrix rhusiopathiae*.

Route of infection: They are normal inhabitants of tonsils and lymphoid organs. The bacillus is

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also present in and around pig sty. Hence infection easily takes place by ingestion of contaminated food and water.

Symptoms: In acute form fever, conjunctivitis, prostration, anorexia, constipation and then diarrhoea, swollen and painful knee joint are the general symptoms. Death may be sudden. In urticarial form (milder than acute) erythematous (reddish) patches on the ventral surface of the body and between hind legs (hence the name ; Diamond skin disease) followed by necrosis and sloughing. Chronic form is also seen.

Control: Vaccine is available which confers immunity for 4-6 months.

Antiserum may be given when the epidemic prevails.

Hygienic upkeep of sties and good sanitation.

Treatment of affected animals

VIRAL DISEASES OF SWINE

Classical swine fever (Hog cholera)

It is an acute generalised infection of pigs caused by a virus and characterised by high fever, prostration and sudden death.

Mode of infection: (1) direct contact with diseased animals (2) through contaminated food and water.

Incubation period: 5-10 days.

Symptoms: The disease can occur in acute or chronic forms. In acute form high fever, loss of appetite, conjunctivitis, vomiting, diarrhoea, stiffness of body, weakness of hind quarters and hemorrhage (purplish discolouration) in the abdomen, ears, snout etc. are noticed. In chronic form, cough, difficulty in breathing and diarrhoea are noticed.

Control: Common measures for controlling contagious diseases. Vaccination of a susceptible population is the most effective way of control. Antiserum is used for treatment in outbreaks.

African swine fever (ASF)

It is a highly infectious and severe hemorrhagic disease of domestic and wild pigs caused by a virus. It produces a wide range of clinical signs and lesions that can closely resemble those of classical swine fever.

Mode of infection: The disease spreads through direct contact with infected pigs, feces or body fluids, indirect contact via fomites such as equipment, vehicles or people who work with pigs, between pig farms with ineffective biosecurity, pigs eating infected meat or meat products and

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through biological vectors (ticks)

The practice of feeding uncooked food waste (that has not been appropriately heat treated) to pigs can also result in the transmission of the virus if the food waste being fed to pigs contains contaminated pork products.

Symptoms: Peracute, acute, subacute, and chronic forms of ASF occur depending on the virulence of the virus. . The acute and peracute forms of the disease are characterized by a severe hemorrhagic disease with close to 100% mortality. After a short incubation period of 3 –7 days, swine develop high fever (up to 42°C), followed by inappetence, incoordination, prostration and death. Surviving swine will then exhibit reddening or cyanosis of the ears and snout, generalized reddening of the body and bleeding from the nose and anus. Infected pigs may also show dyspnea, vomiting, and abortion if pregnant.

Diagnosis: Diagnostic confusion occurs with agents that cause similar clinical signs, such as classical swine fever (hog cholera).

Control: By eradication. The strategy involves ; slaughter of infected swine and swine in contact, safe disposal of carcasses, enhanced biosecurity in backyard farms, prohibiting the feeding of waste food, spraying of acaricides etc.

DISEASES OF GOATS

BACTERIAL DISEASES OF GOATS

Anthrax

Sudden fever and death. Tarry black discharges from the natural orifices. Vaccination and proper disposal of the carcass.

Haemorrhagic septicaemia

Fever and pneumonia. Usually seen during rainy seasons. Vaccinate before the onset of monsoon

Brucellosis

Brucella organisms infect goat's placenta and udder, causing abortion and mastitis. Abortion in the last trimester of pregnancy, infertility, orchitis and arthritis are the common signs.

Diagnosis :The diagnosis can be done by the isolation of organisms and by serological tests.

Control: The test and disposal procedure is highly desirable.

Johne's disease

Johne's disease is a specific chronic contagious enteritis of cattle, sheep, goat, buffaloes and

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occasionally of pigs. The disease is characterized by progressive emaciation, and in cattle and buffaloes by chronic diarrhea and thickening of the intestine.

Transmission: Under natural conditions the disease spreads by ingestion of feed and water contaminated by the faeces of infected animals. Affected animals may not show clinical symptoms and continue to discharge organisms in faeces. The organisms persist in pastures for about a year.

Tetanus

This is an infectious disease of animals and man, and is characterized by spasmodic tetany and hyperaesthesia.

Aetiology: *Clostridium tetani*

Transmission : Infection occurs by contamination of wounds. Deep puncture wounds provide favourable conditions for the spores to germinate, multiply and produce toxin which is subsequently absorbed .

Symptoms : The incubation period is generally 1-2 weeks but it may be as short as 3 days. Tetanus affects many species of domesticated animals but occurs particularly in horses and lambs; less frequently in adult sheep, goats, cattle, pigs, dogs and cats; and rarely in poultry. The initial symptoms are mild stiffness and difficulty to move. More severe symptoms develop after 12-24 hours which are stiffness of limbs, neck, head, tail and twitching of muscles. The spasms develop in response to noise. In terminal stages ears are erect, nostrils dilated, nictitating membrane protruded, progressive paralysis (saw horse). Mastication becomes very difficult because the mouth cannot be opened, hence the name lockjaw. Death occurs due to respiratory failure.

Control: The treatment is done by administering antitoxin and then treating the wound. Proper hygiene and cleanliness should be observed while doing castration and other surgical procedures.

Vaccination as per the schedule is the ideal method for prevention.

Enterotoxaemia

Enterotoxemia, also known as overeating or pulpy kidney disease, is a condition caused by *Clostridium perfringens* type D. These bacteria are normally found in the soil and as part of the

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normal microflora in the gastrointestinal tract of healthy sheep and goats. Under specific conditions, these bacteria can rapidly reproduce in the animal's intestine, producing large quantities of toxins. Young animals are most susceptible. Sudden and high mortality rates may occasionally occur in lambs and kids. Although adult animals are also susceptible to enterotoxemia, they develop immunity due to frequent exposure to low doses of these toxins.

Causes: Excessive consumption of milk, feed with high concentrations of grains and low in roughage, when natural immunity is compromised by diseases, while recovering from an illness or stressed, when animals are heavily parasitized with gastrointestinal parasites, including nematodes, cestodes (tapeworms) and coccidia and when the motility of the gastrointestinal tract is reduced

Symptoms: Typical clinical signs include; Loss of appetite, abdominal discomfort, profuse and/or watery diarrhea that may be blood tinged. The peracute form is most frequent in young animals. It is characterized by sudden death that occurs approximately 12 hours after the first signs of the disease appear. Some kids may show signs of central nervous disease, such as excitement or convulsions.

Diagnosis: based on clinical signs, history of sudden death and confirmation by necropsy examination.

By laboratory Examination

Treatment: *Clostridium perfringens* C & D antitoxin, antibiotic and supportive therapy

Prevention & control: Effective vaccines are commercially available to prevent enterotoxemia. All animals (especially young animals) within the herd should be vaccinated.

DISEASES DUE TO MYCOPLASMA

CCPP(Contagious caprine pleuropneumonia)

Mycoplasma sp. produce a variety of clinical signs in infected small ruminants.

Aetiology: *Contagious caprine pleuropneumonia*.

Mode of transmission: CCPP is transmitted via aerosol droplets

These organisms cause severe pneumonia outbreaks in goat herds with high morbidity and mortality. High fever, signs of respiratory disease accompanied by polyarthritis, mastitis, neurologic signs, diarrhoea, and sudden death, abortion of pregnant goats etc. are major clinical signs of *Mycoplasma* infection.

Diagnosis: Laboratory tests (bacterial culture or PCR assay)

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Treatment and Control: Treatment relies on appropriate antimicrobial administration.

Accurate diagnosis, strict biosecurity, and prompt treatment are essential for decreasing disease transmission and herd losses

VIRAL DISEASES OF GOATS

PPR (Peste des petits ruminants) or Goat plague

It is a viral disease affecting goats causing heavy mortality.

Etiology: It is closely related to the Rinderpest virus.

Incubation period: may range from 2-6 days in field conditions.

Mode of transmission: is by close contact, and confinement seems to favor outbreaks. Secretions and excretions of sick animals are the sources of infection. Transmission can occur during the incubation period. It is generally accepted that there is no carrier state.

Symptoms: The acute form of PPR is accompanied by a sudden rise in body temperature to 104°–106°F Affected animals appear ill and restless and have a dull coat, dry muzzle, congested mucous membranes, and decreased appetite. Early, the nasal discharge is serous; later, it becomes mucopurulent and gives a putrid odor to the breath. Small areas of necrosis may be observed on the mucous membrane on the floor of the nasal cavity. The conjunctivae are frequently congested, and the medial canthus may exhibit a small degree of crusting. Some affected animals develop a profuse catarrhal conjunctivitis with matting of the eyelids. Necrotic stomatitis affects the lower lip and gum and the gumline of the incisor teeth; in more severe cases, it may involve the dental pad, palate, cheeks and their papillae, and the tongue. Diarrhea may be profuse and accompanied by dehydration and emaciation; hypothermia and death follow, usually after 5–10 days. Bronchopneumonia, characterized by coughing, may develop at late stages of the disease. Pregnant animals may abort. Morbidity and mortality rates are higher in young animals than in adults.

Control & Treatment:

Immediate isolation of affected goats from clinically healthy goats is the most important measure in controlling the spread of infection. Nutritious soft, moist, palatable diet should be given to the affected goats.

Measures should be taken for notification of disease to the nearest government veterinary hospital. Carcasses of affected goats should be burned or buried. Proper disposal of contact fomites and decontamination is must. Vaccination is the most effective way to control PPR

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No specific treatment is recommended for PPR being a viral disease. However, mortality rates can be reduced by the use of drugs that control the bacterial and parasitic complications. Lesions around the eyes, nostrils and mouth should be cleaned twice daily with sterile cotton swabs. Fluid therapy, antimicrobials, mouthwash with 5% boro-glycerine can be of benefit in reducing the mortality during outbreak of PPR in goats.

FMD

Fever and lesions in the mouth and foot. Prevention is by vaccination

Goat pox

Low grade fever and pox lesions on the lips and oral mucosa, axilla, groins, nose, udder and teat, underside of the tail and mouth. Control methods as in the case of cattle.

OTHER DISEASES OF GOATS

Foot Rot

Infection in the interdigital spaces and coronary band. It can be controlled by housing the animals in clean and dry houses.

Mastitis

Udder swelling, induration and occasional gangrene. Control methods as in the case of cattle

Pneumonia

Usually caused by bacteria, viruses, mycoplasma etc. Symptoms are fever, cough and respiratory distress. It can be controlled by vaccination, sanitation, proper ventilation, providing clean drinking water etc.

DISEASES OF POULTRY

BACTERIAL DISEASES OF POULTRY

Fowl cholera (Avian Pasteurellosis)

It is an infectious disease of poultry caused by *Pasteurella multocida* causing heavy mortality.

Symptoms: In acute cases birds may die without any signs. In less severe cases affected birds are weak, droopy, open beak breathing, comb and wattles become cyanotic, ruffled feathers, yellowish diarrhoea are other symptoms. Death may occur suddenly.

Diagnosis: From the symptoms, Laboratory tests

Control: Strict hygienic and sanitary measures in and around poultry houses and restriction to

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visitors. Fowl cholera vaccines are available for prevention.

Salmonellosis (Pullorum Diseases)

Salmonellosis is a group of diseases affecting animals, man and birds (poultry) caused by different species of *Salmonella* bacteria.

Pullorum disease is a contagious disease characterised by very high mortality affecting young chicks and turkeys caused by *Salmonella pullorum*.

Mode of transmission

1. Through contaminated food and water.
2. Through eggs of affected birds (vertical transmission).

Vertical transmission is of grave concern. The chicks hatched out from the infected eggs become carriers of the infection. The brooder and incubator are also getting infected. Sometimes the whole farm may be infected in this way and cause heavy economic losses.

Clinical signs: Affected birds huddle near the heat source, are anorectic, weak, depressed, and have white fecal material pasted to the vent area. Other signs include gasping, blindness, swollen joints and lameness.

Diagnosis: By lab tests (Serological test-Rapid plate agglutination test)

Prevention and Control: No treatment is likely to affect. Elimination of carrier birds. Ensure birds/hatching eggs are procured from stock free of *Salmonella* on introduction

Fowl typhoid

It is an infectious disease of chicks and poults caused by *Salmonella gallinarum*

Symptoms: Similar to pullorum disease. Birds show diarrhoea, greenish faeces and systemic disturbances

Diagnosis: By lab tests

Prevention and Control: Treatment is never recommended. Control measures are based on elimination of the birds

Colibacillosis

It is an infectious disease of poultry caused by a pathogenic strain of *Escherichia coli*

In acute forms symptoms resemble fowl cholera or fowl typhoid. Signs vary and can include acute fatal septicemia, airsacculitis etc.

Diagnosis: By laboratory tests

Prevention and Control: Most bacterial isolates are resistant to multiple antibiotics, so

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prevention of exposure through good management is recommended.

Infectious coryza

It is an acute respiratory disease of poultry caused by the bacterium *Avibacterium paragallinarum*.

Clinical signs: Affects upper respiratory tract, seromucoid nasal and ocular discharge, sneezing, and facial oedema.

Diagnosis: By laboratory tests

Prevention and Control: Prevention is based on sound management practices, including appropriate biosecurity measures and vaccination. Early antibiotic treatment may help infected birds recover.

DISEASES CAUSED BY MYCOPLASMA IN POULTRY

Chronic Respiratory Disease (CRD)

Chronic respiratory disease (CRD) is seen in chickens and turkeys which cause major damage to the respiratory system but mortality is low.

Aetiology : *Mycoplasma gallisepticum*.

Mode of infection: In poultry the infection is spread vertically through infected eggs, horizontally by close contact via airborne transmission and indirect or mechanical routes.

Symptoms: The signs of CRD are conjunctivitis, sneezing, shaking of the head, coughing, swelling of the orbital sinuses and tracheal rales, weight loss, reduction in egg production and fertility particularly in turkeys and game birds.

Treatment: There are antibiotics that can help to control the disease but reducing stress is an essential component for managing the disease once it is in the flock.

Prevention and Control: Management issues must be addressed before the birds arrive.

Sanitation and hygiene. Ensure birds/hatching eggs are procured from stock free of *M. Gallisepticum* on introduction.

VIRAL DISEASES OF POULTRY

Ranikhet disease(Newcastle disease)

Malayalam: കോഴി വസന്ത

It is an acute infectious and contagious disease of fowls caused by a virus, characterised by fever, respiratory distress and high mortality.

Incubation period: 3 – 7 days

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Mode of infection: (1) through contaminated food and water. (2) Contact with diseased birds and contaminated utensils

Symptoms: The disease runs for a short period (1-3 days). Mortality rate is 90-100%. Affected birds show loss of appetite and appear sleepy with the eyes partially or fully closed. They cough, sneeze and gasp for breath by half opened beaks. There is mucous discharge from the mouth and nostrils. The droppings will be watery, yellowish white and foul smelling. Nervous symptoms like torticollis, weakness of legs, paralysis of wings etc. are also noticed.

Control: Isolation and disinfection

Immunisation by RDF1/Lasota vaccine and RDVK/R₂B vaccine and Killed vaccine.

Vaccination failure i.e. vaccinated birds getting the disease, has been observed. This is mainly due to non-observance of the following: -

1. Improper deworming or failure to deworm before vaccination.
2. Faulty technique of vaccination.
3. Improper storage of vaccine (failed cold chain)

Avian influenza (Bird Flu)

Malayalam: പക്ഷിപ്പനി

Avian influenza refers to the infection of birds with avian influenza viruses. These viruses occur naturally among wild aquatic birds worldwide and can infect domestic poultry, other bird and animal species.

Mode of infection: Infected birds can shed avian influenza A viruses in their saliva, nasal secretions, and feces. Susceptible birds become infected when they have contact with the virus as it is shed by infected birds. They also can become infected through contact with surfaces that are contaminated with virus from infected birds.

Clinical signs: Affected birds show high fever, oculonasal discharge, cyanosis of comb and wattle, oedema of head and neck, respiratory distress, diarrhoea, drop in egg production etc. Ducks may remain asymptomatic.

Avian influenza outbreaks are of concern in domesticated birds for several reasons:

1. The potential for rapid spread and significant illness and death among poultry during outbreaks of highly pathogenic Avian influenza
2. The economic impact and trade restrictions from a highly pathogenic avian influenza outbreak
3. The possibility that avian influenza Type A viruses could be transmitted to humans

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Control: In case of Avian influenza outbreaks in poultry, depopulation (also called “stamping out”), burial, decontamination of infected flocks and restocking is usually carried out as per the Government of India guidelines. Disposal of dead birds is done by deep burial or incineration. In addition, disease surveillance in domestic and migratory birds should be done.

Fowl pox

Malayalam: കോഴി വസൂരി

Fowlpox is a slow-spreading viral infection of chickens and turkeys characterized by proliferative lesions in the skin that progress to thick scabs (cutaneous form) and by lesions in the upper GI and respiratory tracts (diphtheritic form). Virulent strains may cause lesions in the internal organs (systemic form). The nodules or eruptions appear on the comb, wattle, eyelids and sometimes in the eye, mouth and nasal cavities. Incubation period is 6-15 days. The affected birds show decrease in egg production, difficulty in respiration, off-feed and death can occur by starvation.

Control & Prevention : Immunisation using Fowl pox vaccine. Separation of affected birds, disinfection and individual care and nursing of affected birds are to be done.

Infectious bronchitis

It is the most contagious viral respiratory disease in poultry. Young birds are more susceptible.

Aetiology: Infectious bronchitis virus is a coronavirus

Mode of transmission: Contagious and spreads rapidly by aerosol. Contaminated faeces, litter and fomites spread the virus.

Clinical signs: Sneezing and watery eyes, facial oedema are seen early on, followed by depression, coughing and nasal discharge. Poor egg shell quality, watery albumen, ruffled feathers and wet droppings are seen in laying birds. A drop in egg production and weight gain with deterioration of internal and external quality of egg, tracheal rales, gasping and diarrhoea are also seen.

Diagnosis: From respiratory signs and postmortem lesions.

Virus isolation and PCR are necessary for a definitive diagnosis.

Prevention & control: Prophylactic vaccination

Infectious bursal disease

A viral disease, seen worldwide, which affects the bursa (component of the immune system of

Notes on Disease and Disease control – Livestock Inspectors Training

birds). The damages caused to the immune system by the virus predisposes the birds to other infections. Usually seen in chicks of 3-6 weeks of age. Morbidity is high.

Mode of infection: The route of infection is usually oral, but may be via the conjunctiva or respiratory tract, with an incubation period of 2-3 days.

Symptoms: Usually seen as acute disease in chicks. Common signs include depression, inappetance, ruffled feathers, unsteady gait, huddling under equipment, vent pecking, diarrhoea with urates in mucus (soiled vent).

Diagnosis: From clinical signs, postmortem lesions and laboratory tests

Treatment: No specific treatment

Prevention: Vaccination and biosecurity measures

Marek's disease

Marek disease is a highly contagious lymphomatous and neuropathic viral disease of poultry. MD can occur at any time beginning at 3-4 wks of age or older.

Mode of infection: Direct or indirect contact. Its transmissible through exposure to feathers, dander and body secretions, including dropping

Incubation period: The incubation period can vary, however it generally takes 4 -12 weeks for the disease to develop after the birds have been exposed to the virus.

Symptoms: There are two main forms of Marek's disease; **classical form and acute form**

The **classical Marek's disease** where the Paralysis of the legs and/or wings (with one leg stretched forward and the other held behind) are prominent. The neck is twisted (torticollis), the toes are inwardly curved and the birds show incoordination while walking.

In **the acute Marek's disease** mortality rate is high. The birds may die suddenly, the visceral organs are enlarged, particularly the ovary. The eyes are also affected and they appear grayish white in colour, impaired vision or blindness. Depression and lethargy including pale droopy comb, irregularity around feather follicles, skin looks reddened, rough, thickened or lumpy.

Control: By immunisation of day old chicks

Avian Leukosis (Big liver disease)

Lymphoid leukosis is a neoplastic disease of poultry caused by a virus. The disease is characterized by lymphoma, occurring in chickens approximately 16 weeks of age and older.

Mode of infection: through the egg (vertical) and also by direct contact (horizontal)

Incubation period: 4-6 months

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Symptoms: Anaemia and weakness are the common signs. Internal organs like spleen, liver (most often affected), bursa etc. are enlarged and the abdomen becomes distended. Severe anaemia and weakness lead to the death of the birds. The bones are also enlarged.

Control: Selection of birds from disease free stock. Hygiene and sanitation.

Should be differentiated from MD from following features.

Develop only in chicks above 16-20 wks of age. Uniform involvement of bursa.

Absence of nervous, skin, eye or muscular form of the disease.

FUNGAL DISEASES OF POULTRY

Aspergillosis(Brooder pneumonia/ Mycotic pneumonia)

Aspergillosis is most often a respiratory infection in chickens and turkeys. Severe outbreaks usually occur in **birds 7 – 40 days old**. Sick birds may display respiratory distress, suppressed growth, and general unthriftiness.

Aetiology: *Aspergillus fumigatus*, a mold is the common cause of aspergillosis.

Mode of infection: High mortality rates can be seen in chicks and poults that inhale large numbers of spores during hatching or when placed on bedding contaminated with mold spores. In older birds, infection is caused primarily by inhalation of spore-laden dust from contaminated litter, feed, or dusty range areas. In facilities with recurrent infection, the air handling system(s) should also be investigated as a source of contamination.

Clinical signs: The lungs and air sacs are most frequently involved. Signs include dyspnea, labored breathing, fever, inappetence, and emaciation. Less frequently, a neurologic form might present, with clinical signs that include torticollis and tremors.

Diagnosis: From clinical signs and postmortem lesions. Confirmation is by laboratory tests.

Prevention and treatment: Clean and disinfect the hatchery on a regular basis. Use clean, mold-free litter and feed. There is no treatment available for commercial flocks. Cull the affected and remove contaminated litter. Disinfect the premises before stocking new birds.

DISEASES OF DUCKS

BACTERIAL DISEASES OF DUCKS

Rimerellosis (*Riemerella anatipestifer* Infection/*Pasteurella anatipestifer* infection, infectious serositis, New Duck disease)

This bacterial disease causes high mortality, weight loss and condemnation.

Symptoms: In the acute form, listlessness, eye discharge and diarrhea are commonly seen.

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Ducks show incoordination, shaking of the head and twisted neck. Birds are commonly found on their backs, paddling their legs. Typical lesions found in dead birds are infected air sacs, pericardium, meninges etc.

Control: Preventive management and vaccination are effective means of control. Antibiotics and chemotherapeutic agents are effective in reducing mortality.

Duck cholera (Duck Pasteurellosis)

It is caused by the bacterium *Pasteurella multocida*. It's an important disease of domestic ducks, and is an especially troublesome disease of ducks in some parts of Asia. This disease is associated with poor sanitation, and stagnant water in duck pens.

Symptoms: include loss of appetite, mucous discharge from the mouth, diarrhea, and in breeder ducks, labored breathing.

Control: Good sanitation practices go a long way toward preventing this disease. Chemotherapeutic agents given along with feed are effective in controlling the disease.

Colibacillosis

This common infection of ducks is caused by the bacteria *Escherichia coli*. The common signs are reduced hatchability and infection of the yolk sac (omphalitis); septicemia in ducks aged about 2-8 weeks ; salpingitis and peritonitis in breeder ducks. In market ducks, *E. coli* infection produces lesions very similar to those seen in *Riemerella anatipestifer* infection.

Control: Good sanitation and management are important preventive measures.

VIRAL DISEASES OF DUCKS

Duck plague(Duck Virus Enteritis)

Duck plague is a highly contagious infectious disease of ducks, geese, and swans caused by a virus. Non-waterfowl bird species are believed to be resistant to the virus. Duck plague outbreaks in domestic waterfowl can result in mortality rates as high as 60% and can cause significant decreases in egg production that can lead to major economic losses.

Transmission: Susceptible birds can become infected with duck plague by coming into contact with a bird that is shedding the virus, or by coming into contact with an environment that is contaminated with feces or oral and nasal secretions from an infected bird. Ingesting food or water contaminated with the virus is another common mode of transmission. Susceptible birds may become infected by way of inhalation of aerosolized secretions containing the virus. The virus can also be passed in eggs from a female to her offspring. The virus can remain viable in

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the environment for weeks or even longer under certain environmental conditions (up to 60 days). Stress makes the birds more susceptible to disease and the overcrowding facilitates disease transmission.

Clinical Signs: In the field, mortality is commonly the first sign of disease as sick birds often die before signs of illness are observed. Clinical signs can vary depending on the viral strain, and the species affected, age, sex, and immune status of the infected bird. Clinical signs may develop within 3-7 days of exposure to the virus and include depression, ruffled feathers, difficulty breathing, loss of coordination, photophobia, extreme thirst, loss of appetite, ocular and nasal discharge (that may be blood tinged), blood tinged and watery diarrhea, and a blood tinged vent. Females may experience decreased egg production, while males may have a prolapsed penis. Some birds will survive and become asymptomatic carriers of the virus and can act as sources for future outbreaks, particularly in times of stress. These birds often have a “cold sore” like ulcer under the tongue from which the virus is shed.

Diagnosis: A presumptive diagnosis can be made based on characteristic lesions seen on necropsy. Laboratory testing is required for a definitive diagnosis.

Treatment: As with most viral diseases, there is no treatment for duck plague.

Prevention & Control : Prompt carcass disposal and environmental decontamination is key in control of duck plague outbreaks as the virus is very persistent in the environment. When outbreaks occur in domestic flocks, the entire flock is often destroyed to prevent surviving birds from carrying the virus and initiating future outbreaks. The premises must also be decontaminated to the extent possible and the carcasses burned. Vaccines against duck plague are used in the commercial duck industry. Domestic birds should be prevented from coming into contact with wild birds.

Duck virus hepatitis

Duck virus hepatitis is a highly fatal contagious disease of young ducklings (1-28 days of age). Ducklings are most susceptible at the younger ages and gradually become more resistant as they grow older. The disease is rarely seen in ducklings over 4 weeks of age. The onset of the disease is very rapid, it spreads quickly through the flock and may cause up to 90% mortality. Sick ducklings develop spasmodic contractions of their legs and die within an hour in a typical "arched-backward" position. The liver is enlarged and shows hemorrhagic spots.

Prevention and control: Keep the age groups isolated and vaccinate breeder ducks (to

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produce maternally immune ducklings).

FUNGAL DISEASES OF DUCKS

Aspergillosis

This condition occurs when ducks inhale spores produced by the mold (fungi) *Aspergillus*. (*Aspergillus fumigatus* is the common species) that grows on damp straw or feed. These inhaled spores cause multiple nodules or plaques in the lungs and air sacs.

Symptoms: Common signs include gasping, listlessness and dehydration.

Control: The best solution to prevent aspergillosis is to avoid using moldy straw and preventing feed from getting wet.

VACCINATION IN DOMESTIC ANIMALS AND BIRDS

An ideal vaccine should confer prolonged and strong immunity in vaccinated animals and induce rapid onset of immunity. It should preferably stimulate responses distinguishable from those due to natural infection so that vaccination and eradication may proceed simultaneously. The risks of vaccination must not exceed those caused by the disease itself. There are different types of vaccines like; inactivated vaccine, live attenuated vaccine, subunit vaccine, toxoid, viral vector vaccine, mRNA vaccine, conjugate vaccine etc.

Points to be considered before administering vaccines

- Animals should be in good health at the time of vaccination. Do not vaccinate animals which are already in stress (bad weather, scarcity of fodder & water, disease outbreaks, after transportation etc.)
- Deworm the animals & remove ecto-parasites one to two weeks prior to vaccination.
- Do not vaccinate animals 60 days before slaughter
- Do not give antibiotics or immunosuppressive drugs like corticosteroids to animals being vaccinated.
- Transportation of vaccines should be made as fast as possible and the cold chain should be maintained throughout until it is used.
- For the drinking water method of vaccination in poultry; Waterers should be thoroughly cleaned before use, do not use any disinfectant, use clean and cool water, do not use

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chlorinated water, withhold water for at least 2 hrs so that the birds become thirsty and provide ample water space. Mix only the required quantity of vaccine as it should be used within 2 hrs.

- Multidose vials of vaccines should be used within the specified time after opening. Prolonged storage of such vials after opening is not desirable.
- Vaccine should be reconstituted in the prescribed diluent and administered in proper site and route
- Strictly follow manufacturers instructions, vaccination schedule
- Vaccination should be done during the cool hours of the day.
- In rare cases hypersensitivity may occur, then immediate treatment with antihistamines may be done.
- Left out animals; young ones below 3 months, diseased ones, malnourished, advanced pregnancy etc. should be vaccinated as soon as they are fit for it
- Keep records of vaccination including name of the vaccine, manufacturing company, Batch number, Expiry date, Dose & route of vaccine.
 - Create a stress free environment for animals after vaccination.
 - Properly dispose of used vaccine vials and unused balance vaccines
 - Milking animals may have a temporary reduction in yield for one or two days

VACCINATION SCHEDULE FOR CATTLE

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Sl. No.	Disease	Vaccine	Age at first dose	Dose and Route	Booster dose	Remarks
1	Foot and Mouth Disease (FMD)	FMD Oil adjuvant vaccine	4 months and above	Dose varies between manufacturers i/m	Six monthly	Pregnant animals above 7 months should be avoided
2	Haemorrhagic Septicaemia (HS)	HS Oil adjuvant vaccine inactivated	6 months	2.5 ml i/m	Annually	Repeat yearly before monsoon
3	Black Quarter	BQ vaccine inactivated	6 months	5 ml s/c	Annually	Repeat yearly before monsoon
4	Brucellosis	B.abortus Strain 19 vaccine	4-8 months of age(Only female calves)		Once in a lifetime	Only serologically negative female calves should be vaccinated. Bulls and pregnant animals should not be vaccinated.

VACCINATION SCHEDULE FOR GOATS

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Sl. No.	Disease	Vaccine	Age at first dose	Dose & Route	Booster Dose	Remarks
1.	Anthrax	Anthrax spore vaccine - LIVE	4-6 months	0.2 ml s/c at tail fold	Annual	Only in endemic areas
2.	Hemorrhagic Septicemia (H.S.)	HS Oil adjuvant vaccine - inactivated	3-4 months	2ml i/m	Annual vaccination	Before monsoon
3.	Enterotoxaemia	Enterotoxaemia vaccine inactivated	3-4 months	2.5 ml s/c	After 2 weeks Repeat annually	Follow manufacturer's instructions
4.	Black Quarter (B.Q)	BQ vaccine inactivated	4 months	5 ml s/c	After 10 days in endemic areas Annual vaccination recommended	”
5.	P.P.R.	PPR Ruminants Live tissue culture vaccine	3 months	1 ml s/c at neck region	Repeat every three year	”
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Note: In goats regular vaccination is recommended for FMD, Enterotoxaemia and PPR. In endemic areas vaccination against HS, Anthrax and Tetanus is advised.

VACCINATION SCHEDULE FOR PIGS

Sl. No	DISEASE	Name of Vaccine	Age at First Dose	Dose & Route	Booster	REMARKS
1	Swine erysipelas	Swine erysipelas vaccine	3-4 weeks	Repeat 3-6 weeks later. Repeat every 6-9 m	1 ml s/c	Follow manufacturer's instructions
2	Swine fever	Classical Swine Fever Lapinised vaccine – Live attenuated	2 weeks in dam is not vaccinated 2 months if dam is vaccinated	Annual	1 ml i/m	”
3	FMD	FMD Oil adjuvant vaccine	3-4 months	Repeat every 6-9 months	As per manufacturer's recommendation	”
4	Tetanus	Tetanus toxoid vaccine	4-6 weeks	12 weeks Annual	1 – 2 ml s/c or i/m	Usually done prior to surgery and wound treatment

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VACCINATION SCHEDULE FOR BROILERS

Sl.N o.	Disease	Vaccine	Age at first dose	Dose & Route	Remarks
1	Marek's Disease	Marek's Disease Vaccine (HVT)	Day 1 (day of hatch)	0.2 ml s/c in the back side lower neck region or i/m	If the birds are to be kept for more than 60 days
2	Ranikhet Disease	RDF/La Sota	5-7 days	Nasal/ocular one drop each	
3	IBD	IBD Georgia strain	14 th day	Drinking water	Add good quality ice @ 100 gm/litre of water, milk powder @ 6 gm/litre of water
4	Ranikhet Disease	La Sota/RDV	21 st day	Drinking water	
5	IBD	IBD Intermediate strain	28 th day	Drinking water	Add good quality ice @ 100 gm/litre of water, milk powder @ 6 gm/litre of water

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<u>VACCINATION SCHEDULE FOR LAYERS</u>					
Sl. No.	Disease	Vaccine	Age at first dose	Dose & Route	Remarks
1	Marek's Disease	Marek's Disease Vaccine (HVT)	day 1 (day old)	0.2 ml s/c in the back side lower neck region or i/m	As per manufacturer's instruction
2	Ranikhet Disease	Ranikhet Disease Vaccine- RDF/La Sota	5-7 days	Oculo nasal one drop each	”
3	Infectious Bursal Disease Vaccine	IBD Vaccine - LIVE	10-14 days	Drinking water	Add good quality ice @ 100 gm/litre of water, milk powder @ 6gm/litre of water
4	Ranikhet Disease	Ranikhet Disease Vaccine La Sota/F	21 days	Drinking water	
5	IBD	IBD – LIVE vaccine	28 th day	Drinking water	Add good quality ice @ 100 gm/litre of water, milk powder @ 6 gm/litre of water

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VACCINATION SCHEDULE FOR COMMERCIAL DUCKS

Sl.No	Disease	Name of Vaccine	Age	Dose & Route of administration	Booster Dose	Remarks
1	Duck plague	DPV Live attenuated	6 weeks	s/c wing web	12 weeks Then annually	Annual booster
2	Duck pasteurellosis	Duck pasteurella (Oil adjuvant - inactivated)	4 th week – 0.3 ml i/m in leg muscle. Booster after 30 days in leg or breast muscle Repeat 0.5ml i/m every six months	Ducks below 6 months – 0.3ml i/m Ducks above 6 months – 0.5 ml i/m		Repeat after 6 months
3	Duck Viral Hepatitis	Duck Viral Hepatitis vaccine	2-4 weeks 8 weeks 16 th weeks	i/m or s/c i/m or s/c i/m or s/c/ foot web	May be repeated at 4-5 months interval	

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LIST OF VACCINE MANUFACTURERS

1. M/s Indian Immunologicals Ltd, Rakshapuram PO, Hyderabad
 2. IVP, Ranipet, Tamil Nadu
 3. IAH&VB, Palode, Thiruvananthapuram, Kerala
 4. IAH&VB, Hebbal, Bangalore
 5. IVRI, Izatnagar, UP
- ATS and toxoid available from
Serum Institute, Pune; Halffkine Institute, Bombay; King Institute, Madras

VACCINES PRODUCED IN VBI

INSTITUTE OF ANIMAL HEALTH AND VETERINARY BIOLOGICALS, PALODE DETAILS OF VIRAL VACCINE PRODUCED

Name of vaccine	Presentation		Reconstitution and Diluent	Dose and Route	Recommended Age of vaccination	Immunity	Storage	Price Rs GST 5% extra
	Packaging	Doses						
Ranikhet Disease Vaccine (F) Live attenuated	2 ml vial	100 doses	PBS 10 ml	1 drop intraocular 1 drop intranasal	On day of Hatch to 7 days	6-8 Weeks	12 months (20°C)	35.00
Ranikhet Disease Vaccine (Kornorov/ Mukteshwar) Live attenuated	2 ml vial	200 doses	PBS 100 ml	0.5 ml S/C wing web	Six to Eight weeks Booster dose at 16 th to 18 th week	One productive year	12 months (20°C)	70.00
Duck Plague Vaccine Live attenuated	2 ml vial	200 doses	PBS 100 ml	0.5 ml S/C wing web	1 st dose – 6 th week Booster – 12 th week then once in an year	1 year	12 months (20°C)	80.00
Fowl Pox Vaccine Live attenuated	2 ml vial	100 doses	50% Glycerol Saline 1 ml	Prick method- 2 pricks 1 cm apart in wing web	Six to eight weeks of age	1 year	24 months (20°C)	40.00
Classical Swine Fever Vaccine Live attenuated	2 ml Vial	10/ 15 doses	Normal Saline or PBS 10/ 15 ml	1ml i/m	2 week if dam is not vaccinated Two months if dam is vaccinated Then once in an year	One year	6 months (20°C)	750.00/ 15 doses
Peste des Petits Ruminants Cell Culture Vaccine Live attenuated	2 ml Vial	50 doses	Normal Saline or 1 M MgSO ₄ 50 ml	1ml S/c at neck region	3 months	More than 3 years	12 months (20°C)	400.00

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INSTITUTE OF ANIMAL HEALTH AND VETERINARY BIOLOGICALS, PALODE DETAILS OF BACTERIAL VACCINE PRODUCED

Name of vaccine	Presentation		Reconstitution and Diluent	Dose and Route	Recommended Age of vaccination	Immunity	Storage	Price Rs GST 5% extra
	Packaging	Doses						
Anthrax spore vaccine – Live	100 ml Bottle	100 doses	Administered as such	Cattle - 1 ml s/c at neck Goats - 0.2 ml s/c at tail fold Sheep - 0.5 ml s/c at neck region Pig – 0.5 ml s/c at inner thigh Elephant – s/c at caudal fold 20 yrs and above – 3ml 15-20 yrs - 2.5 ml 10-15 yrs - 2 ml 5-10 yrs - 1.5 ml 2-5 yrs - 1 ml	First vaccination in animals above 6 months of age Annual vaccination recommended in endemic areas	1 year	24months (' 4 °C)	445.00
Black Quarter Vaccine - Inactivated	300 ml Bottle	60 doses	Administered as such	Cattle - weighing less than 140 kg - 5ml s/c - weighing more than 140 kg – 10 ml s/c Sheep and Goat – 5ml s/c	First vaccination in animals above 4 months of age Booster after 10 days in endemic areas Annual vaccination recommended	1 year	24months (' 4 °C)	170.00
Hemorrhagic Septicaemia (Oil Adjuvant) Vaccine - Inactivated	300 ml Bottle	150 doses	Administered as such	Cattle - weighing less than 140 kg - 2ml i/m - weighing more than 140 kg – 3ml i/m Sheep and Goat -2ml i/m	First vaccination at 4 to 6 months of age Annual vaccination to be done 1 month before monsoon	9 months	12months (' 4 °C)	1165.00
Duck Pasteurella (Oil Adjuvant) Vaccine - Inactivated	100 ml Bottle	200 doses	Administered as such	Ducks below 6 months – 0.3 ml Above 6 months -0.5 ml i/m	0.3 ml i/m at 4 th week in leg muscle Booster 0.3 ml after 30 days in leg or breast muscle Repeat 0.5 ml i/m every 6 months	6 months	12 months (+4°C)	335.00
Enterotoxaemia Vaccine – Inactivated	100 ml Bottle	40 doses	Administered as such	Kids above 3 months of age – 2.5 ml s/c at neck	Above 3 months - 2.5 ml Booster after two weeks Repeat every 12 months	12 months	12 months (+4°C)	560.00

COMMON PARASITIC INFESTATIONS IN ANIMALS

The parasites are living organisms, which live at the expense of another living organism and cause damage to that organism. The organism in which the parasite lives is called a host. If the animal harbors the adult stage of the parasite, it is called the final or definitive host and if the host harbors the immature or larval stage of the parasite it is called the intermediate host.

Certain parasites live inside the body of the host in different tissues; organs, blood etc. and they are called endoparasites. Ex: Roundworms, Tapeworm, Flukes, and Protozoa. Certain others live on the surface of the body of the host and are called ectoparasites. Ex: Flies, Fleas, Ticks, Mites etc. Majority of parasites produce eggs when they reach the adult stage and from the eggs, the larvae develop and grow to maturity. When the larval stage of a parasite develops in the ground or other surrounding, the parasite is said to have a direct life cycle.

The stage of the parasite, which enters the final host (cattle, sheep, goat, dog, etc.) and sets up infection to the host is called infective stage. The eggs or larvae can infect a host. The common modes are (1) by ingestion along with food and water e.g. Eggs of roundworms, cysts of coccidia, larvae of flukes etc. (2) through pastures while grazing eg. Larvae of flukes of cattle, goats, and sheep. (3) By penetration of skin or mucous membrane eg. Larvae of blood flukes (4) direct contact eg. Infection with mites. (5) By the bite of arthropods like ticks flies,

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mosquitoes eg. Protozoan infections.

The parasites produce harmful effects in the body of the host in different ways (1) by absorbing the food materials from the intestine of the host (roundworms and tapeworms). (2) By sucking the blood (hookworms, ticks). (3) By causing mechanical obstruction to intestinal lumen, (Large Roundworms) bile ducts (Liver fluke), Blood vessels (Blood flukes). (4) By producing harmful wounds on the skin through which other pathogenic organisms may enter (Flies, ticks, mites). Hookworm produces wounds on intestinal mucous membranes. (5) By causing damage to the important organs like liver, lungs etc. (liver fluke, lungworms) (6) By producing harmful substances in the body of the host (gastro- intestinal worms). (7) By transmitting other pathogenic organisms like bacteria, viruses etc. (flies, ticks). (8) By reducing the resistance of the host and thereby leading to other diseases.

The parasitic infestations can be diagnosed from symptoms and by laboratory examination of materials like faeces, blood, nasal discharges, skin scraping etc.

Parasitic infestations can be controlled by (1.) Avoiding contamination of food and water with excreta of animals and birds. (2.) Proper hygiene and sanitary measures. (3) destruction of intermediate hosts, which carry the larval stages of parasite. (4.) By maintaining the proper health of the animals by proper attention to feeding. (5.) Treatment of affected animals. One of the most common reasons for anthelmintic resistance is improper or sub therapeutic medication. Rotate the group of drugs every year to prevent anthelmintic resistance and always administer optimum/therapeutic dose.

Parasites may be multicellular or unicellular. Roundworms, tapeworms etc. are multicellular. Protozoa are unicellular organisms and can be seen only by a microscope, which cause diseases directly or transmit other disease-producing organisms to animals and birds .The parasites are grouped mainly as

1. Roundworms (nematodes)

Elongated and cylindrical worm with pointed ends. There are male and female roundworms. The female worm produces eggs and passes out in faeces or other discharges. They vary in size and the large roundworms may measure 25-30 cm or more. The roundworms seen in animals and birds parasitise the gastrointestinal tract. They are also seen in organs like the liver, lungs, heart ,etc. Most roundworms have a direct life cycle.

2. Tape worms (cestodes):

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Flat tape like or ribbon like worms with a head and a body formed of numerous segments. Each segment contains the male and female reproductive organs and is capable of production of eggs. The segments at the posterior end of the worm when they contain fully matured eggs are called ripe or gravid segments. Gravid segments get detached from the worm and passed out in faeces. The tapeworms are seen in the intestinal tract. They have an indirect life cycle and various arthropods act as intermediate hosts.

3. Flukes:

Flatworms or leaf-like or conical in shape. It has only one segment for the body and it contains the male and female reproductive organs. The eggs produced by flukes passed out in faeces or other discharges. The flukes are seen in liver, (liver fluke), rumen and reticulum (ruminal fluke) blood (blood flukes) etc. They have an indirect life cycle and various types of snails act as intermediate hosts.

4. Protozoa:

These are unicellular organisms and can be seen only under a microscope. They are seen in the blood cells, cells of the intestine etc. They have a direct or indirect life cycle. Ticks and biting flies transmit many protozoan diseases. Eg. Babesia, Theileria, Coccidia

5. Insects and ectoparasites:

They are collectively called arthropods (having paired and jointed legs). There are male and female arthropods. All arthropods are ectoparasites. The body is divided into distinct parts like the head, thorax and abdomen. Most of the insects have wings (fleas, mosquitoes). The mosquitoes, flies, fleas, lice, ticks and mites are included in this group.

Diseases Due To Roundworms.

Ascariasis in animals.

(Infestation with large roundworms *Toxocara vitulorum*)

Affects cattle, buffaloes, pigs and dogs. The young animals are mainly affected. In cattle and buffaloes the large roundworm is found in the small intestine. The female worms after fertilization by males lay large numbers of eggs, which pass out along with dung. The larva develops in the egg and ingestion of food and water contaminated by such eggs sets up infection in animals. From the eggs that are swallowed by the animals the larvae come out in the intestine and these larvae pierces the intestinal wall and reach the liver. From the liver they go to the heart, lungs, trachea, pharynx, and then swallowed back and finally reaches the intestine. By this time

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the larvae grow to the adult stage. (8-9 weeks after infection.) The larva in the liver, lungs, and heart causes severe damage to these organs. Affected calves usually develop pneumonia, cough and become weak and debilitated. Common symptoms are lack of appetite, dullness, diarrhoea, anaemia, stunted growth, stomach pain and extreme weakness.

In dogs the large roundworm causing ascariasis are *Toxocara canis*. They produce similar harmful effects seen as in ascariasis of cattle. The young pups become weak, anaemic, stunted in growth and show distended abdomen.

In pigs the large roundworm is *Ascaris suum*. The harmful effect produced by this worm is also similar to other large roundworms.

Diagnosis: by examining the faeces and from symptoms.

Treatment: Piperazine compounds at a dose of 200—300 mg/ kg. body weight is effective against these worms.

Albendazole dose for ruminants is 5 mg/kg body weight

Fenbendazole dose for ruminants is 7.5-10 mg/kg body weight.

Control: Proper disposal of manure, cleanliness and good management of young animals, avoiding damp and dirty places, treatment of affected animals etc. is important. Pregnant animals may be dewormed before parturition. Calves and puppies have the habit of eating sand and licking various objects. This should be prevented.

Strongylosis In Ruminants

(Infestation with strongyle worms)

Strongyle worms are slender and medium sized and measure about 3.0 cm in length. They infect all ruminants and are seen in the stomach, intestine and rarely in lungs. Many of them have teeth like structures at their anterior and by which they injure the intestinal mucosa and feed on blood. The anterior end of certain worms is bent like a hook and are called hookworms. The female strongyle worms after fertilization produce eggs and are passed out along with faeces. The larvae escape from the eggs and undergo some development in the outside and become the infective stage. Animals get infected by ingestion of these larvae from the pasture or contaminated food and water. The larvae of hookworms can penetrate the skin and enter the body of the host. In the body of the host these larvae develop into adults in the predilection sites.

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Stomach worm infestation:

The stomach worms injure the mucosa and suck blood. Their larvae cause minute hemorrhages in the mucosa of the intestine. Anemia, oedematous swelling of the inter mandibular space (bottle jaw), diarrhoea, loss of appetite and weakness are the common symptoms. Young animals are severely affected and may cause death.

Nodular worm: affects cattle, sheep, goats and other ruminants, as well as pigs. The developing larval stages of nodular worms cause nodules in the mucosa of the large intestine (caecum and colon) which is called as pimply gut or knotty gut and hence called "nodular worms". The intestine becomes rigid and the peristalsis is affected. Secondary infection of nodules causes the formation of abscesses. Persistent diarrhoea, weakness and emaciation are the clinical symptoms. The hookworms are voracious bloodsuckers and produce severe anaemia, weakness and diarrhoea.

Lungworm infestation: Parasitic Bronchitis, Husk, Hoose.

The lungworms cause severe inflammation of bronchi (bronchitis), which extends to lungs leading to verminous pneumonia or "Husk". Cough, mucous exudate from nostrils, difficulty in breathing and rapid respiration are the clinical symptoms.

Treatment And Control: Proper feeding, regular treatment, avoiding wet and moist places and contamination of food and water, rotational grazing, avoiding overcrowding, proper hygiene etc. are the common control measures. Albendazole (7.5-10 mg./kg. body weight), Fenbendazole (5 mg/kg. body weight) is effective. Fenbendazole is safe for use in pregnancy.

Ancylostomiasis In Dogs

(Infestation with hookworms.)

The major hookworm of dogs is *Ancylostoma caninum* and is seen in the small intestine. They have teeth at the anterior end by which they injure the mucosa of the intestine and feed on the blood of the host. The infective larva enters the body of the host by penetration of skin and they wander through heart, lungs, etc., before they become adults. Prenatal infection is also possible. Dogs of all ages are affected and they show anemia, oedema, weakness and emaciation. The faeces is diarrhoeic and dark brownish in colour with mucus and blood.

Treatment & control: Albendazole (12.5-30 mg/Kg body weight) and Fenbendazole (50 mg/Kg daily for 3 consecutive days) are effective.

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Round Worms Of Poultry

Large roundworm : *Ascaridia galli*, the large round worm of poultry seen in the small intestine. The worm causes enteritis, anaemia diarrhoea and the affected birds show decreased egg production and emaciation.

Piperazine compounds are effective.(Margin of safety is low)

< 6wks age: 50 mg/kg

> 6 wks age: 100 - 200 mg/kg

Syngamus trachea is a strongyle worm of poultry and commonly known as “gapeworm”. They are seen in the trachea. The male and female worms always remain attached to each other. The worms cause inflammation of the trachea and may obstruct the passage leading to respiratory difficulty. The affected birds gape or gap for air by stretching the head and keeping the mouth open. They may sneeze, cough and shake their head frequently. These worms also suck blood and cause anaemia. Fenbendazole (20 mg per Kg body weight for 3 consecutive days) is effective in treating this infection. Periodical oral administration of garlic extract will help to reduce the incidence of infection.

Control measures are similar to other roundworms.

Diseases Due To Tapeworms:

Tapeworm infestation in ruminants (Moniezia)

They are seen in the small intestine and their gravid segments are passed out in dung. These segments appear like cooked rice grains. Certain small mites seen on the grass (grass mites) eat the gravid segments and the larvae develop in them. When the animals grazing on pastures ingest these mites containing larval forms of the tapeworms, they get infected. Young animals are more affected and they become stunted in growth and show anemia, oedema, diarrhoea, rough coat and weakness.

Praziquantel (5-10 mg per Kg body weight),

Niclosamide (70 -100 mg per Kg body weight) ,

Oxyclosanide(10-15 mg/Kg), (Closantel(10 mg/Kg), Rafoxanide (7.5 mg per Kg body weight), and Ivermectin suspension is effective.

Control measures are directed against destruction of the grass mites on the pasture by ploughing and planting suitable crops.

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Tapeworm Infestations in Dogs

(Taeniasis in dogs)

Dipylidium caninum is the common tapeworm of dogs. Dogs get infection by accidental ingestion of these intermediate hosts. These larval stages, especially the larvae of *Echinococcus* (Hydatid cyst), cause serious damage to many of the internal organs of the intermediate hosts. The dogs lose condition due to depletion of food by the adult tapeworms and become emaciated. Regular treatment of affected dogs is essential since man and other domestic animals are affected by the larval stages of tapeworms.

Praziquantel 5-10 mg/Kg is effective against the worms

Tapeworm infestations in poultry

The larvae develop in ants and slugs. Birds get infection by ingestion of these intermediate hosts. Niclosamide (100-150 mg/Kg), Praziquantel (10 mg/kg body weight) can be used for treatment.

Control should be directed against the intermediate hosts. Insecticide may be used for destroying flies and ants.

Larvae of human tapeworms (Taenia solium and Taenia saginata) are seen in pigs and cattle causing harmful effects.

Diseases Due To Flukes:

Liver flukes: *Fasciola hepatica* and *Fasciola gigantica* are the flukes found in the bile duct of cattle, sheep, and goats. They are leaf-like, large flukes. The larval stages of these flukes undergo development in water snails. Animals get infection by ingesting these larval forms along with plants or grass blades.

These worms cause damage to the liver and the disease is known as liver rot, clay pipe cirrhosis or fluke cirrhosis. The symptoms noticed are emaciation, anaemia, diarrhoea, oedema of intermandibular region (bottle jaw) and anorexia.

Rafoxanide (7.5 - 10 mg per Kg body wt.) & Oxcyclozanide (10-15 mg per Kg body wt.) are effective.

Ruminal Flukes (Amphistomes): The Amphistomes are thick and fleshy flukes, conical in appearance and infect cattle, sheep and goats. These flukes are seen in the rumen and reticulum. The eggs passed by these flukes develop in the same way as that of liver flukes and snails act as intermediate hosts. Infection in animals is similar to liver flukes.

The developing flukes in the intestine produce irritation, and cause diarrhoea. Chronic diarrhoea,

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debility and bottle jaw are the common symptoms. The disease is called Immature Amphistomiasis or Pitto.

Oxyclosanide 15 mg per kg body weight is effective (margin of safety is low).

Blood Flukes (Schistosomiasis): They are found in the blood vessels. The common blood fluke is *Schistosoma nasalis*, seen in the blood vessels of nostrils of cattle and buffaloes, causing a disease called Nasal Schistosomiasis, nasal granuloma or “snoring disease. The infective larva of blood flukes can enter the body of animals by penetration through the skin, when they drink water from infected pounds. Snails are the intermediate hosts.

The symptoms are difficulty in breathing and a snoring sound due to the formation of cauliflower like growths in the nostrils. The animals become very weak and emaciated. Effective treatment is available.

Schistosoma spindale seen in hepatic and mesenteric veins of cattle and causes dysentery. The infective larvae enter the body of the host orally or through penetration of the skin.

Treatment: Drugs effective for fluke infection include Praziquantel (20 mg/kg body weight), Rafoxanide (15 mg/kg) and Oxyclozanide(18 mg/Kg) is also effective.

Prevention and control: Fluke infestation can be prevented by the control and destruction of the intermediate hosts, which are always snails of different species. The important point to remember is to repeat the treatment after monsoon in endemic areas. Destruction of snails by using copper sulphate should be done every year in endemic areas. The vegetation in ponds and other places from where the animals are watered should be cleaned so that the breeding of snails is prevented. Snails should be removed by hand picking or by allowing ducks to feed. Chemicals like copper sulphate (1 part in 5 million) are used to kill snails in streams, ponds etc. In moist pasture lands copper sulphate is mixed with sand and spread at the rate of 10-30 lbs. per hectare.

DEWORMING SCHEDULE OF CATTLE, GOATS, PIGS, DOGS & POULTRY

Deworming recommendations

Ruminants

- In calves, broad spectrum anthelmintic (eg. Pyrantel pamoate) should be given within 2 weeks of age (10 to 16 days of age).
- In pregnant cattle, anthelmintic treatment in the third trimester of pregnancy or within 10 days of parturition reduces the risk of transplacental/ transmammary/ periparturient egg rise.
- Pregnant does may be dewormed 3-5 weeks prior to kidding depending on climatic factors.
- Deworming may be practiced twice a year during the onset of south-west (July) and north-east monsoon(October) in organised farms.
- In goats it is not suggested to follow a specific deworming schedule. In field condition it is preferred to give a broad-spectrum anthelmintic to kids at 2-3 weeks of age. Further deworming is needed only in clinical cases. Adult animals may be dosed as per Anaemia Eye Card/ Body Condition Score only. In conditions where faecal sample examination is not possible, deworming can be done at six months interval with a broad spectrum anthelmintic active against cestodes also. Piperazine shall not be administered to goats. Anticoccidials should also be administered during the first month of age, preferably after faecal sample examination.
- During drier parts of the year/ summer months, animals need not be given anthelmintics unless clinically affected.
- Monitor anthelmintic resistance annually in the farm. World Association for the Advancement of Veterinary Parasitology has recommended Fecal Egg Count Reduction Test (FECRT) and Egg Hatch Assay (EHA) as standard tests for field detection of resistance to anthelminmtics. Regular monitoring of resistance with FECRT/ EHA is recommended in farms and if resistance is detected, stop the use of the anthelmintic and change over to another drug with different mode of action.

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- To avoid introducing resistant worms into a farm, quarantine animals brought from an area where resistance has been detected, immediately deworm them with at least two drug classes and keep them away from the pasture for at least 30 h.
- Fasting 12 to 24 h prior to treatment or 2 treatments given 12 h apart is also beneficial.
- Deworming should be done in confinement, keeping the animals away from grazing areas for two days to avoid contamination of grazing lands.
- Under-dosing and unnecessary treatment may be avoided.

Pigs

The deworming schedule should include pre breeding for all breeding stock and pre farrowing for gilts and sows, prevention of Strongyloides and roundworms in piglets, and one or more dewormings in weanling and growing pigs.

- Boars - every 6 months
- Sows - 2 weeks before farrowing and after weaning
- Piglets - 1 week after weaning
- Fatteners - 1 week after weaning and 3 months later
- Gilts - 1 week after weaning, 3 months and at 7 months of age at least 2 weeks before service

Dogs and cats

- Pups may be dewormed at 2, 4, 6 and 8 weeks of age
- From 3 months onwards, monthly faecal sample examination may be done to monitor nematode, cestode and trematode infections and treated accordingly.
- In kennels, where puppy mortality due to toxocarosis is a persistent threat the following regimen may be recommended.

Pregnant female dogs: Fenbendazole @ 50 mg/kg b.w. daily from 40th day of pregnancy until 14th day post-partum

- Weaned pups and kittens brought home should be dewormed for a minimum of 3 treatments at 2 weeks interval.

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Poultry

Poultry flocks can pick up a variety of different parasites, both external and internal. Small infestations may not cause any health problems, but a heavy infestation can reduce performance, animal welfare and can sometimes lead to death. It is important to monitor the flock for any signs of parasites. *Ascaridia galli*, *Heterakis gallinae*, *Syngamus trachea*, *Tetrameres* spp. and *Capillaria* spp. are the common roundworms whereas *Railletina*, *Davainea*, *Amoebotaenia*, *Choanotaenia* etc. are the common tapeworms of poultry. Under deep litter system and other floor system of rearing, deworming may be required. However, in cage system deworming has to be carried out only after faecal sample examination for the presence of ova of parasites.

Under deep litter system and other floor system of rearing, deworming should be started at 5-6 weeks of age and repeated at 1-2 months interval. In case of cage system, deworming should be given only after examining the EPG (egg per gram of droppings). A particular group of anthelmintic should be given for the maximum of three consecutive times to avoid drug resistance. The dewormer medicated water should be drunk by the birds within 4 hours.

1. Piperazine @ 50mg/kg for <6wks old chicks, 100mg/kg for >6wks old chicks and 200mg/kg for adult birds with repetition after 7-10 days is effective against roundworms.
2. Albendazole @ 5mg/kg for single day is effective against *Ascaridia galli*, *Heterakis gallinarum* and *Capillaria obsignata*. It is effective against cestodes at the dose rate of 20mg/kg.
3. Flubendazole @ 1.43mg/kg is effective against *A. galli* and *H. gallinarum*
4. Cambendazole @ 60mg/kg for 3 days is effective against *Ascaridia anseris* and at 30mg/kg against *Trichostrongylus tenuis*.
5. Mebendazole @ 10mg/kg for 3 days is effective against *Amidostomum anseris*, *Trichostrongylus tenuis* and *Dispharynx nasuta*.
6. Fenbendazole @ 10-50mg/kg for 3-5 days is effective against *Ascaris* spp., *Capillaria* spp and *Syngamus trachea* (gapeworm). Toxic to pigeons, if used at the rate of 30mg/kg for 5 days.
7. Thiabendazole @ 75mg/kg controls *T. tenuis*.
8. Tetramisole @ 40mg/kg is effective against *A. galli*, *H. gallinarum* and *C. obsignata*.
9. Pyrantel tartrate @ 15-25mg/kg is effective against *A. galli* and *Capillaria* spp. and 100mg/kg is effective against *A. anseris* and *T. tenuis*.

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10. Citarin @ 40mg/kg is effective against *A. anseris* and *T. tenuis*.
11. Levamisole at 25-30mg/kg is effective against *A. dissimilis*, *H. gallinarum* and *C. obsignata*.
12. Phenothiazine @ 100-500mg is used to treat *H. gallinarum* (caecal worm) in chicken and black head disease in turkey which is transmitted by *H. gallinarum*.

For tapeworm infestation, ordinary dewormers will not remove the scolex. Butynorate with piperazine and phenothiazine shows some efficacy. The promising drugs are chlorophene, niclosamide and praziquantel.

13. Hexachlorophene at 25-50mg/kg is highly efficient against *Raillietina cesticillus* in chicken
14. Niclosomide (cestoNex-N) 175mg/kg is effective against tapeworms. It eliminates not only segments of tapeworms but also the scolex. It is also active against trematodes.
15. Praziquantel 10-20mg/kg s/c or oral or i/m with repetition after 10 days is effective to control tapeworms. 10mg/kg for 14 days to control trematodes.
16. Manson's eyeworm (*Oxyuris mansoni*) can be treated by applying local anesthetic in the eye and expose the lacrimal sac by lifting nictitating membrane and apply 1-2 drops of 5% cresol solution in the lacrimal sac to kill the worms immediately. After killing irrigate the eye with sterile water and eyes will become clear after 2-3 days.

PROTOZOAN DISEASES IN CATTLE

Trichomoniasis

T. foetus lives on the surface of the penis and prepuce of the bull and in the reproductive tract of the cow. There are no obvious signs of Trichomoniasis in the male, and pregnancy loss is the only sign of the disease in the female.

Transmission: occurs during natural breeding. A bull can infect a cow and vice versa.. However, most infected cows eventually clear the infection. Once a bull is infected, it remains so for life. Therefore, most control programs focus primarily on the detection and elimination of infected bulls. After clearing the infection, cows can rebreed and carry a fetus to term. The period of immunity is short and will not protect subsequent pregnancies if the cow is re-exposed to an infected bull.

Prevention and control: All newly acquired cows that are less than 120 days pregnant should be isolated from the breeding herd. They may be placed in the breeding herd once they are four months pregnant.

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Coccidiosis

It is a protozoan disease affecting animals and birds. They live and grow within the cells lining the gastrointestinal tracts of the host. The oocyst passed out along with faeces undergoes some development in the ground (sporulation) and may contaminate food and water thereby setting up infection in other animals and birds. Young animals like calves, lambs, and kids are usually affected. After a susceptible goat ingests sporulated oocyst, "spores" are released and enter the cells lining the intestine. In the intestine they go through several stages of development. The intestinal cells are destroyed and thousands of smaller forms of coccidia are released. The disease is also known as Red Dysentery, which is more common in calves.

Symptoms: Range from loss of appetite, short-lived diarrhea containing blood and mucus to severe cases involving straining, large quantities of dark and blood tinged faeces, dehydration, anaemia progressive emaciation and death.

Control: When an outbreak occurs, isolation and sanitation are key to preventing its spread throughout the herd. Additionally, coccidiosis is more of a problem in crowded conditions. Care should be taken to check dehydration. Proper hygiene and optimum management of young animals is essential for controlling the disease.

Chemotherapeutic agents such as sulpha drugs and amprolium are effective in treating coccidiosis. Sulphaquinoxaline (15 mg/kg for 3 to 5 days) and (Amprolium 10 mg/kg for 5-7 days) etc. are effective .

Babesiosis (Piroplasmosis, Red water fever)

It's a protozoan disease caused by an organism called *Babesia* and transmitted by ticks. This disease affects cattle, horses, sheep, goats, swine, dogs and cats. There are different species causing the disease in different animal species. These protozoa are seen in pairs in the red blood corpuscles.

Clinical signs: Fever, coffee coloured urine (due to the excretion of free haemoglobin liberated from RBC - hemoglobinuria) , progressive anemia, jaundice and death in severe cases.

Diagnosis: From symptoms and Laboratory tests

Control and Treatment: Tick control via acaricides or management practices can be useful in reducing tick burden, which can lower transmission rates. Diminazene aceturate is effective against *Babesia*.

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Theileriosis

It's a protozoan disease caused by *Theileria* in cattle and transmitted by ticks. The organisms are found in RBC's and WBC's.

Clinical symptoms: high fever, swelling of superficial lymph glands, dyspnea and high mortality.

Control: Spraying or dipping of animals with acaracides is the most frequently used method for prevention of theileriosis and vaccination in endemic areas. Effective vaccines are available.

Clinical cases can be treated with appropriate chemotherapeutic agents.

PROTOZOAN DISEASES IN POULTRY

Coccidiosis in poultry

Coccidiosis is caused by the protozoa of family Eimeriidae causing heavy mortality. There are several species of coccidia affecting birds. The infectious process is rapid (4 - 7 days) and is characterized by parasite replication in host cells with extensive damage to the intestinal mucosa. The disease spreads rapidly and the chicks die in large numbers.

Mode of infection: Coccidia are almost universally present in poultry-raising operations, but clinical disease occurs only after ingestion of relatively large numbers of sporulated oocysts by susceptible birds. Both clinically infected and recovered birds shed oocysts in their droppings, which contaminate feed, dust, water, litter, and soil. Oocysts may be transmitted by mechanical carriers (eg. equipment, clothing, insects, farm workers, and other animals).

Clinical signs : Major symptoms are dullness, stunted growth, crouching , poor performance, crowding at a corner, ruffled feathers, severe blood tinged diarrhoea, the chicks show signs of acute pain and high mortality. Survivors of severe infections recover in 10–14 days but may never recover lost performance.

Diagnosis: Coccidial infections are readily confirmed by demonstration of oocysts in feces or intestinal scrapings of dead birds

Control: Coccidiosis in chicks and growers can be prevented by incorporating suitable coccidiostats in feed. The coccidiostats used in feed may be rotated every three to six months to prevent the parasites developing resistance. Strict sanitation, good litter management and adequate floor space will help to check the occurrence

Amprolium compounds can be used for the prevention and cure.(Narrow margin of safety).

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For treatment Amprolium powder 20% - 1 gm per litre of drinking water for 7 days.

During treatment medicated drinking water should be the only source of drinking water.

DISEASES DUE TO INSECTS AND ECTOPARASITES.

Flies: The blowflies or bazaar flies (commonly called blue bottle fly and green bottle fly.) and house flies (musca) lay eggs on the wounds in the skin.

Cutaneous myiasis(Maggot wound)

In summer the flies are a menace to the animals since they cause cutaneous myiasis, which is common in Kerala. This is due to the fact that the flies come and dispose of their eggs on minor cuts and wounds. The eggs hatch out and the niggers feed on blood and cellular debris of the wound. The maggots burrow deep in the wound and cause cellular and vascular damage causing blood loss. So the fly-borne wounds should be attended to by dressing the wound. Clean the wound, remove accessible maggots and apply a larvicide (Turpentine oil , eucalyptus oil, larvicidal sprays etc.). Apply soothing dressing once the wound is free of maggots.

Prevention: By sanitation; Manure and other decaying plant and animal material should be promptly disposed of.

Mechanical control using sticky traps and ultraviolet light traps placed around the shed reduce housefly populations. Flies breed in and around cattle sheds and urine pits etc. These areas should be periodically cleaned and lime should be put in such places, to prevent fly breeding.

Lice: The lice develop all the stages in the body of the host and are so called as permanent parasites. Their eggs are called Nit. Infestation with lice is called Lousiness. Due to the irritation caused by the lice on the body, the animals and birds become restless and feeding is affected. Milk production may be decreased. Cattle may lick the body to relieve the irritation and during this process, they may swallow a lot of hair resulting in formation of 'hairballs' in the stomach.

Ectoparasiticides like Pyrethroids, Ivermectin, Amitraz, etc. can be used for the treatment .

Ticks: are blood-sucking arthropods and transmit protozoan diseases like Babesiosis and Theileriosis. The larvae of the ticks are called 'seed ticks'. The tick infestation can be controlled by dips and sprays containing permethrin, amitraz, spot on/ pour on preparations containing fipronil and selamectin .

Mites: are very minute ectoparasites and cause a disease called Mange or Scabies in animals. The affected skin becomes thickened and severe itching and scab formation are symptoms. Sarcoptes and Psoroptes are the common mites in cattle, sheep and goats. In dogs Sarcoptes and

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Demodex cause mange. Benzyl benzoate lotion, selamectin, Ivermectin etc. are effective in treating mange.

Fleas: The dog flea causes irritation to the skin and it also transmits the dog tapeworm.

Permethrin and other pyrethroid compounds are effective in controlling fleas.

Treatment and Control of Ectoparasites.

The following practice may be made to control ticks

1. An acaricide canal around the cattle shed should be made, when the buildings are made, so that the ticks are not having access from nearest vegetation to the shed
2. An acaricide should be used to destroy tick populations in newly introduced animals before they are put in with other animals.
3. Vegetation around cattle sheds should be periodically cleaned and burned.
4. Rodent population in the farm should be controlled. They act as reservoirs to the ticks.
5. Dipping, spraying, dusting, etc. should be periodically practiced for ectoparasites and changed according to the species and body sites of affection.

For flies, fleas, lice, ticks and mites - Deltamethrin 1-6 ml per litre of water

For fleas, lice, ticks and mites - Flumethrin pour on

For ticks and mites - Amitraz 5 ml per litre of water

ZOONOTIC DISEASES

Zoonotic disease, also called zoonosis, is a group of diseases that can be transmitted to humans by nonhuman vertebrate animals, such as mammals, birds, reptiles, amphibians, and fish. Humans can contract zoonotic diseases through direct contact with infected animals, and also by consumption of contaminated food or water, inhalation, arthropod vectors (such as flies, ticks, and mosquitoes) and pests.

1. BACTERIAL ZOONOSES

1. Anthrax

Anthrax can be transferred from infected/dead animals to humans by contact, inhalation of spores, eating undercooked infected meat, by handling animal products like hides, environmental contamination by spores, by flies etc. There are three forms of anthrax infection in human beings: cutaneous (skin-carbuncles), inhalation (lungs) and gastrointestinal (stomach and intestine) forms.

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2. Brucellosis (undulant fever/malta fever)

Humans contract the disease by direct or indirect contact with infected animals, infected foetal membranes, foetus, placenta etc. or animal products contaminated with the bacteria. Brucellosis causes a range of symptoms, some of which may be present for prolonged periods of time.

3. Tuberculosis

Zoonotic TB is a form of tuberculosis in people caused by *Mycobacterium bovis*, which belongs to the *M.tuberculosis complex*. The zoonotic form is primarily transmitted indirectly, through the consumption of contaminated milk, dairy products and meat. Another mode of transmission is by inhalation.

TB usually affects the lungs, but it can also affect other parts of the body, such as the brain, the kidneys, spine etc..

4. Salmonellosis

Salmonellosis is a food-borne infection typically caused by consumption of improperly cooked food, such as meat, eggs or unpasteurized milk/dairy products. People can also get salmonellosis by direct contact with faeces from infected animals or ingestion of contaminated food and water and mechanically by arthropods. People with salmonellosis may have diarrhea, fever and abdominal cramping. Different species of *Salmonella* can cause illness.

Prevention:

Proper cooking of food and its storage are important. Prevent contamination of stored food. Infected persons should not handle food. Ensure meat products from Salmonella free sources.

5. Listeriosis

Listeria monocytogenes causes listeriosis, a bacterial disease in ruminants and humans. Most infections in people occur by eating raw meat or unpasteurized dairy products (**oral**), but there are additional ways (inhalation, direct contact etc.) to be infected. The disease primarily affects pregnant women, newborns, older adults, and people with weakened immune systems. Symptoms in humans include fever, muscle aches, headache, stiff neck, confusion, loss of balance, and convulsions.

6. Plague(Yersiniosis)

Plague is an infectious disease caused by the bacteria *Yersinia pestis*, a zoonotic bacteria, usually found in small mammals and their fleas. Humans can be infected through the bite of infected

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vector fleas, wound infection and droplet infection .

Patients develop fever, chills, extreme weakness, abdominal pain, shock, and possibly bleeding into the skin and other organs. Skin and other tissues especially on fingers, toes, and the nose may turn black and death occurs (black death).

7. Leptospirosis (Weil's disease) Humans may be infected by direct/indirect contact with urine or reproductive fluids from infected animals, through contact with urine-contaminated water or soil, or by consuming contaminated food or water. The infection route is through abrasions or cuts in the skin, or through the conjunctiva and mucous membranes. Infection also occurs through indirect contact with soil, water, foods, and vegetation contaminated with urine of infected animals and through aerosol from infected animal urine.

Weil's disease can cause a wide range of symptoms. Some infected persons are symptomless. If not treated, Leptospirosis can lead to kidney damage, meningitis (inflammation of the membrane around the brain and spinal cord), liver failure, respiratory distress, and even death. Recently a number of cases have been reported from Kerala in pineapple plantation areas and among NREGA (National Rural Employment Guarantee Scheme) workers.

Wear protective clothing especially gloves if you have contact with infected animals . Always wash your hands after touching animals or anything that may have been contaminated with animal urine.

2. RICKETTSIAL ZOONOSES

1. Q Fever (Coxiellosis)

Q fever results from infection by the bacterium, *Coxiella burnetii*, which can infect people who inhale aerosolized organisms, or by other routes. Most human infections are associated with cattle, sheep and goats, and often occur when the animal gives birth. Symptoms of Q fever include fever, chills, night sweats, headache, fatigue, chest pains. and abortion or premature delivery in pregnant women.

2. Psittacosis (Ornithosis)

Chlamydia psittaci can infect people exposed to infected birds, causing a disease called Psittacosis .It is most commonly associated with pet birds; like parrots and cockatiels, and poultry like; turkeys and ducks. Psittacosis can cause respiratory illness or pneumonia in humans.

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3. VIRAL ZOONOSES

1. Rabies (hydrophobia in man)

Human Beings get rabies from the bite (**direct contact**) of an infected animal, but can also be exposed to the virus by entry of saliva, brain or spinal cord fluid of an infected animal into cuts or breaks in the skin or mucous membranes. Early human symptoms include fever, headache, confusion and abnormal behavior (neurological signs). Once signs begin, recovery is very rare. If you are bitten by an animal, the animal may be tested or quarantined, and contact your physician immediately to determine whether post- exposure treatment is indicated.

2. Avian Influenza

Infected birds shed bird flu virus through their saliva, mucous and feces. Human infections with bird flu viruses can happen after unprotected contact with infected birds or surfaces contaminated with bird flu viruses. The virus gets into a person's eyes, nose or mouth, or is inhaled. The reported signs and symptoms of bird flu virus infections in humans have ranged from no symptoms or mild flu-like upper respiratory symptoms, conjunctivitis, to severe such fever and pneumonia.

3. Contagious Ecthyma (Sore Mouth)

Orf virus infection in animals is commonly referred to as sore mouth, scabby mouth, or contagious ecthyma. Animals infected with orf virus typically develop scabby sores (lesions) around their lips, muzzle, and in their mouth. People become infected by **direct contact** with affected animals. Humans that are infected typically develop ulcerative lesions or nodules on their hands.

4. Japanese encephalitis

A viral encephalitis transmitted to human beings by the bite of infected mosquitoes. Pigs act as amplifier host

5. Kyasanur Forest Disease (KFD)

A viral haemorrhagic fever transmitted to human beings by the bite of ticks

6. Nipah Virus infection

A highly fatal bat-borne viral zoonotic disease in human beings which causes encephalitis and even death.

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7.Cow pox

Most human cases of cowpox appear as one or a small number of pus filled lesions on the hands and face, which then ulcerate and form a black scab before healing on their own.

4. FUNGAL ZOONOSES

1. Ringworm (Dermatophytosis)

Ringworm is a common fungal disease caused by dermatophytes. People can become infected by direct contact with the spores on an infected animal. The spores may be on the animal's hair/wool or skin, and can even be on fomites such as brushes or clippers. Itchiness is the most common symptom, and the spots are generally inflamed at the edge with redness, scaling, and occasionally blistering.

5. PARASITIC ZOONOSES

There are certain parasitic diseases which are transmitted from animals to man. They are

1.Schistosomiasis

Infection occurs by penetration of the skin by the larvae of flukes.

2.Larva migrans:(cutaneous larva migrans & visceral larva migrans)

Cutaneous larva migrans is manifested as an erythematous, serpiginous, pruritic, cutaneous eruption caused by accidental percutaneous penetration and subsequent migration of larvae of various nematode parasites. (*Toxocara* and *Ancylostoma* sp.).

Visceral larva migrans is a systemic manifestation of migration of second stage larvae of nematodes through the tissue of human viscera.

3.Hydatidosis (Cystic echinococcosis)

It is caused by infection with the larval stage of *Echinococcus granulosus*, a small tapeworm found in dogs (definitive host) .Infection is by ingestion of infectious material (eggs of tapeworm). The larval form of the tapeworm may lodge in liver, lungs etc. where they form a fluid-filled sac known as a hydatid cyst

4.Taeniasis

Taeniasis is an infection within the intestines by adult tapeworms belonging to the genus *Taenia* (beef tapeworm & pork tapeworm). When the eggs are consumed by an intermediate host such as a cow or pig, they hatch within the duodenum to become larvae, penetrate through the intestinal wall into nearby blood vessels, and enter the bloodstream. Once they reach organs such as the skeletal muscles, liver or lungs, the larvae then develop into a cyst, a fluid-filled

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cysticercus. These contaminated tissues are then consumed by human beings through raw or undercooked meat. Intermediate stages develop in humans and cattle. The cysts develop in vital organs of man and cause diseases.

METABOLIC DISEASES

METABOLIC DISEASES OF CATTLE

Metabolism is the sum of the physical and chemical processes in an organism by which its material substance is produced, maintained, and destroyed, and by which energy is made available. Metabolic disease or disorder is characterised by disruption of one or more metabolic processes that are connected to the control of a specific metabolite in body fluids.

Production Diseases are diseases affecting high producing cattle resulting from faulty management and nutrition. In high producing animals input of nutrients through feed is less when compared to output of nutrients through milk. So a negative nutrient balance occurs and production diseases are manifested. Examples: hypocalcaemia, ketosis, hypomagnesaemia .

Milk fever

Milk fever is a complex metabolic disease in cows that usually occurs around calving and most commonly in high-productive cows from three years and older. It is one of the most common mineral-related metabolic diseases affecting dairy cows and is also known as calcium deficiency, hypocalcemia, and parturient paresis. A cow with milk fever usually shows very general symptoms of illness which appear when the blood calcium levels are low; lack of appetite, lower body temperature, looks dull, and has difficulties in urination and micturition (also known as subclinical milk fever). If not treated the cow won't get up and will lay down in a typical position on her chest, with her head on her side (known as clinical milk fever)

The clinical signs of milk fever can be classified into three stages: **Stage 1** (Standing, animal reluctance to move or eat and falls easily), **Stage 2** (Sternal Recumbency with lateral kink of neck) **Stage 3** (lateral Recumbency, coma, death occurs due to respiratory failure if untreated)

Calcium shouldn't ever be added before calving as a milk fever preventative measure. Dietary calcium intake should be low as well (about 20 g per day). Give rations low in calcium and higher (relatively) in phosphorus during the dry period of the cow. In this way, the calcium metabolism will be activated. At calving, the parathyroid gland will be better prepared to mobilize calcium from the bones and calcium from ration and supplements will be better

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absorbed. Dry cows should receive sufficient magnesium since hypomagnesemia is highly related to milk fever. Vitamin D is important for the intestinal absorption and bone mobilization of calcium.

Downer cow syndrome

When a cow is unable to stand, it becomes recumbent and frequently referred to as being “down,”. The primary causes of recumbency can be divided into 4 categories; metabolic (eg. hypocal, hypomag), inflammatory (mastitis, metritis), neurological (nerve paralysis) and traumatic (fracture tendon, ligament). When the primary cause is eliminated but the cow still doesn’t rise and has been down for a while, it is referred to as a “downer cow.” The term downer cow denote nonambulatory cattle recumbent for at least 24 hours without obvious reason. Usually they are bright and alert and they eat and drink normally. Occasionally non-alert downers are also seen. Ruling out the known causes of recumbency is required for diagnosis.

Ketosis

Ketosis is a metabolic disorder that occurs in cattle when energy demands (e.g. high milk production) exceed energy intake resulting in a negative energy balance. Ketotic cows often have low blood glucose concentrations. When large amounts of body fat are utilised as an energy source to support production, fat is sometimes mobilised faster than the liver can properly metabolise . If this situation occurs, ketone production exceeds ketone utilisation by the cow, and ketosis results. In the dairy cow, the mismatch between input and output usually occurs in the first few weeks of lactation, because the cow is not able to eat enough to match the energy lost in the milk.

Symptoms

Reduced milk yield, weight loss, reduced appetite, dull coat, acetone smell of breath. Some develop nervous signs including excess salivation, licking, aggression etc.

Control

The initial aim of treatment is to restore the deficiency of glucose in the body. A quick-acting glucose supplement is required immediately. Drenching with propylene glycol or glycerine has long term effects.

Prevention depends on adequate feeding and management practices.

The body condition of the dairy cow is important at calving. Cows should be on a rising plane of nutrition up to calving with the aim to calve in good condition.

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Ruminal acidosis/Acid indigestion

Ruminal acidosis is a metabolic disease of cattle. Acidosis occurs when the pH of the rumen falls to less than 5.0. (normal rumen pH is 6.2 to 7.2; in simple indigestion pH will be within normal limits; in alkaline indigestion it is more than 7)

At first rumen becomes atonic. This depresses appetite and production. Secondly, the change in acidity changes the rumen flora, with acid-producing bacteria taking over. They produce more acid, making the acidosis worse. The increased acid is then absorbed through the rumen wall, causing metabolic acidosis, which in severe cases can lead to shock and death.

Acute acidosis: The primary cause of acidosis is feeding a high level of rapidly digestible carbohydrate such as cereals, often resulting in death, and most commonly seen where cattle have obtained access to excess feed.

Sub-acute acidosis: milder form, is seen as a result of feeding increased concentrates compared to forage.

Symptoms : In Acute acidosis cattle may become depressed, go off feed, have an elevated heart rate or diarrhea. Acute acidosis often results in death.

In sub-acute acidosis common signs are reduced feed intake, dullness, cessation of rumination, feeble rumen motility, poor body condition and weight loss.

Prevention: The key to prevention is reducing the amount of readily fermentable carbohydrate consumed at each meal. This requires both good diet formulation (proper balance of fiber and non fiber carbohydrates). Ruminant diets should also be formulated to provide adequate buffering. This can be accomplished by feed stuff selection and/or by the addition of dietary buffers such as sodium bicarbonate, magnesium oxide etc. Add probiotics, yeast supplements etc to feed which reduces the risk by selectively inhibiting ruminal lactate producers.

METABOLIC DISEASES OF GOATS

LACTIC ACIDOSIS (Grain overload)

Grain overload is an acute disease of small ruminants due to ingestion of excess carbohydrate.

Symptoms: It is characterized by depression, anorexia, bloat, diarrhea, regurgitation of rumen contents through mouth, dehydration, incoordination, collapse, and in severe cases, death.

Diagnosis: is confirmed by rumen pH <5.5 and excessive grain in the diet.

POLIOENCEPHALOMALACIA (PEM)

Polioencephalomalacia is a common metabolic disorder in goats characterized by neuro-muscular alterations due to Thiamine (Vitamin B1) deficiency. Younger animals are more frequently affected than adults because preruminant animals depend on dietary thiamine.

Symptoms: The main clinical signs reflect dysfunction of the cerebrum and include wandering, circling, blindness, incoordination, head pressing, recumbency, nystagmus, and seizures. Some animals are found dead.

Diagnosis: Clinical diagnosis is often difficult and suspected based on the combination of neurologic signs, elimination of other diagnoses, and response to thiamine administration

HYPOCALCAEMIA

Parturient paresis in pregnant and lactating goats is a disturbance of metabolism characterized by acute-onset hypocalcemia and rapid development of hyperexcitability and ataxia, progressing to depression, recumbency, dilated pupil, coma, and death. Unlike parturient paresis in dairy cattle, which primarily occurs within a few days of calving, this condition usually occurs before, and less commonly after parturition in goats. Parturient paresis is due to a decrease in calcium intake under conditions of increased calcium requirements, usually during late gestation. During late gestation there is increased demand for calcium because mineralization of the fetal skeleton occurs 1–3 weeks prepartum, particularly when multiple fetuses are present in utero. This results in a low serum calcium concentration, particularly in animals pregnant with multiple fetuses. Some cases are complicated by concurrent pregnancy toxemia

Control: Prevention is based on provision of adequate dietary calcium throughout gestation.

PREGNANCY TOXEMIA

Pregnancy toxemia, the most common metabolic disorder of pregnant small ruminants, occurs during the final stage of gestation as the result of inappropriate metabolism of carbohydrates and fats. The disease is characterized by hypoglycemic encephalopathy, with a variety of behavioral and neurologic clinical signs.

It mainly affects adult goats having multiple kids. During the last two months, the kids are adding 70% of their birth weight. During the final weeks, there are additional nutritional requirements for the kids as they continue to increase in size and there is less and less room for

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the rumen to hold the same quantity of food. The mother's body will give priority to the kids nutritional needs at her expense. The doe may not be able to consume enough nutrition and the body will start converting stored fats. This leads to the release of ketone bodies into the bloodstream resulting in ketoacidosis.

Symptoms: Loss of appetite, unable to get up or move around, sweet-smelling breath, limping and swelling of feet, muscle tremors, convulsions, blindness, star gazing, recumbency and death

Control: Oral propylene glycol/glycerol. Offer high energy food and seek immediate veterinary assistance.

NONSPECIFIC DISEASES OF CATTLE

COLIC

The term “colic” refers to abdominal pain rather than a specific disorder. Conditions that commonly cause colic include gas, impaction, grain overload, sand ingestion, and parasite infection etc. It's common in horses.

Symptoms: pain, constipation, kicking, groaning. Figure out the cause and treat the colic.

BLOAT

Bloat is simply the build up of gas in the rumen. This gas is produced as part of the normal process of digestion, and is normally lost by belching (eructation). Bloat occurs when belching is prevented.

There are two types of bloat.

The least common type is free gas **bloat**, which occurs when the digestive tract is obstructed (often by foreign objects such as jackfruit, potatoes, tapioca, tumors etc.) or when the animal can't burp (such as with milk fever or tetanus).

The second type of bloat is **frothy bloat**, which happens as the result of a stable foam developing on top of the rumen liquid, which blocks the release of the gas. This is the most common form of bloat, and unlike gassy bloat, it is highly seasonal. This is because the foam is formed by breakdown products from rapidly growing forages (particularly legumes). These increase the viscosity (stickiness) of the rumen fluid and prevent the small bubbles of gas formed by rumen fermentation from coming together to form free gas that can be belched off.

Clinical Signs: Distended left abdomen is the most obvious sign usually associated with respiratory distress, pain, discomfort, and bellowing; death may happen if immediate relief is not given; in some cases sudden death may be the first sign seen.

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Diagnosis: Based on the clinical signs described above and history of access to lush pasture

Control & First Aid:

- In a few cases a trochar and cannula punched through the side into the rumen will relieve gassy bloat when a stomach tube has not worked. But such cases are rare, and as the trochar provides a tremendous opportunity for introduction of infection, it should only be used as a last resort.
- For frothy bloat, antifoaming agents that disperse the foam should be given by stomach tube. Old remedies such as vegetable oil are effective but newer treatments such as dimethicone or simethicone are easier to give as the effective dose is much smaller.
- If an outbreak of frothy bloat occurs all cattle should be put on a high fibre diet (hay or straw), and any cows showing bloating signs treated with an anti-foaming agent.

PNEUMONIA

Pneumonia is a condition seen in animals characterized by “inflammation” of the lung caused by bacteria, virus, fungi, parasite, protozoa, dust, foreign matter etc. Symptoms of bacterial pneumonia include a moist or productive cough, rapid respiratory (breathing) rate, nasal discharge, loud breathing sounds, malaise/depression, loss of appetite, and/or weight loss. Inflammation also stimulates excessive secretion of airway fluid and mucus, resulting in cough and difficulty breathing. The most common causes of pneumonia are infectious in origin.

Aspiration of gastrointestinal contents during vomiting, regurgitation episodes, faulty drenching of medicines etc can cause aspiration pneumonia.

Prevention and control: Proper housing and ventilation, Sanitation.

Antibacterials and anti-inflammatory drugs are used for treatment.

WARTS (Papillomatosis)

Warts are hairless, normally benign growths on skin or mucosa caused by a papillomavirus. Warts can occur anywhere on the body, but they are most frequently seen on surfaces prone to abrasion such as the head (especially around the eyes, ears and lips), neck, shoulder, udder, teats and brisket.

Treatment : Surgical removal, ligation of pedunculated warts and autogenous vaccines have been used to treat warts, but the effectiveness is uncertain because self cure is common.

ANAEMIA

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All farm animals can be affected

Aetiology: Nutritional and as a sequela to various infections

Symptoms: Characterised by general weakness and lack of vigor. Iron deficiency results in non-formation of Haemoglobin

Prevention: Balanced ration. Treatment to be aimed to eliminate the route cause.

HEAT STROKE

Heatstroke is hyperthermia accompanied by a systemic inflammatory response causing multiple organ dysfunction and often death.

Symptoms: include temperature >105.8, dyspnoea, salivation, open mouth respiration, restlessness .

Diagnosis: from clinical signs. First aid includes rapid external cooling by ice pack or cold water irrigation.

NONSPECIFIC DISEASES OF GOATS

UROLITHIASIS IN GOATS

Obstructive urolithiasis (urinary stones) is a significant cause of mortality in male ruminants. Uroliths form from protein and mineral components of the diet. They develop in the urinary bladder and lodge in the distal sigmoid flexure of cattle and small ruminants or the vermiform appendage (urethral process) of the small ruminant, causing obstruction.

Urine supersaturation of mineral components occurs as a result of increased renal excretion, negative water balance, urine pH, and the presence or absence of crystallization inhibitors along with vitamin A deficiency. Substances like suture, tissue debris, blood clots, excess protein, or bacteria in the urine facilitate stone formation.

Early recognition and treatment are critical to achieving a positive outcome. Urolithiasis is a multifactorial disease, making prevention strategies challenging to develop.

BLOAT

Bloat formation occurs when animals consume young leaves and grasses, unknown weeds, easily digestible cereals, rotten vegetables and fruits. This will be followed by diarrhoea, dysentery which may lead to recumbency and death.

Control: measures are the same as in cattle.

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ABORTIONS

1. Nutritional Problems

Proper nutrition is essential for having healthy kids. Simultaneous deficiencies of energy and protein can cause abortion of embryos early in the pregnancy. Deficiencies of some trace minerals such as copper and iodine can be the cause of abortions.

2. Infectious Problems

An abortion in one or more goats in the herd may indicate an infectious disease that needs an overall management response.

- Chlamydiosis - caused by Chlamydia. Abortion typically occurs in the last 2 months of pregnancy and especially the last 2 weeks..
- Toxoplasmosis - this is associated with a coccidium of cats. Cats become infected by consuming uncooked meat scraps, placentas, and small rodents. Goats become infected by eating grass, hay or grains contaminated by cat feces. It can result in abortion, stillbirths and weak kids.
- Q Fever - a bacterial disease capable of being transmitted from animals to humans caused by Coxiella burnetii, a rickettsial organism. C. burnetii may be found in sheep, cattle, goats, cats, dogs, some wild animals (including many wild rodents), birds, and ticks. Animals shed the organism in their urine, feces, milk, and especially in their birth products. Abortion or stillbirths occur in late pregnancy. Placentas and aborted fetuses should be destroyed by burning.
- Listeriosis (Circling disease) - caused by Listeria monocytogenes, an organism commonly found in soil, water, plant litter and digestive tract of ruminants. Abortions occur in the last 2 months of pregnancy.

FOOT ROT

Infection in the interdigital spaces and coronary band. It can be controlled by housing goats in clean and dry houses.

MASTITIS

Udder swelling, induration and occasional gangrene.

Gangrenous mastitis : Inflammation of udder, bluish discoloration of udder and teat skin and the mammary gland becomes cold due to necrosis.

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Control: measures are the same as in cattle.

DISEASES DUE TO TOXINS IN ANIMALS & BIRDS

Toxicology : It is the study of nature and effects of poisonous or toxic substances.

Toxicity : It is the inherent capacity of substance to produce harmful effects.

Poison : It is any substance(liquid/solid/gas) that causes deleterious effects in a living organism.

Venoms : These are substances produced by reptiles for their defense mechanism. E.g. Snake Venom.

Toxins : These are the poisonous substances produced in animal tissues by the action of bacteria or waste products that are not removed from the body due to liver or kidney failure.

Antidote : These are the drugs or chemical agents that neutralize the effects of poison.

Sources of poisoning

There are two major sources of poisoning in animals

- 1) Natural Sources
- 2) Human oriented sources

1) Natural Sources

1.1. Plants : ആനത്തൊട്ടാവടി , പുഷ്പി (**Lantana**), ചേല **Atropa belladonna** (ഉമ്മം) , Strychnos nuxvomica (കാഞ്ഞിരം), Nitrate rich plants, Plants containing HCN (tapioca, rubber and acacia leaves) etc.

1.2. Animals : Poisonous snake bite (Cobra, Krait, Russel viper). Scorpion bite, Toad toxin, Tick toxins and Spider venom.

1.3. Minerals or metals : Arsenic, Lead, Mercury, Selenium, Fluoride.

1.4 Mycotoxins are naturally occurring toxins produced by certain molds (fungi) : Aflatoxins

2.) Human oriented sources

2.1 Accidental causes

a) Fertilizers : Urea, phosphate or Nitrate fertilizers. **b) Insecticides** : Organophosphates, Organochlorines, Carbamates, Pyrethroids. **c) Rodenticides** : Zinc phosphide. **d) Industrial effluents** : Lead, Fluorine, Cyanide, Mercury, Nitrate. **e) Radiation hazards**

2.2 Malicious poisoning

Unlawful discriminial killing of animals by administering poisons e.g. Zinc phosphide, Strychnine, etc.

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Aflatoxin poisoning

Molds (fungi) that grow on cereal grains and oilseeds before and after harvest produce a number of toxins. Wet harvest conditions encourage the growth of this mold. By far the most toxic of these substances is a group of toxins called aflatoxins. Aflatoxins are hepatotoxic. Aflatoxins are produced by the molds *Aspergillus flavus* and *Aspergillus parasiticus*. Ducks are highly susceptible to these toxins. Even small quantities will cause heavy mortality.

COLLECTION, PRESERVATION AND DESPATCH OF CLINICAL MATERIALS FOR LABORATORY EXAMINATION

PURPOSE:

1. Diagnosis of the disease.
2. Confirmation of tentative diagnosis
3. Prognosis
4. To observe the effect of treatment and give direction for future therapy.

GENERAL CONSIDERATIONS FOR COLLECTION OF SPECIMENS:

1. The diseases most commonly encountered in animals are of bacterial, viral, parasitic, fungal and metabolic origin. Diagnosis based on symptoms and laboratory examination of the relevant materials is essential for initiating treatment at the proper time. In general the following points should be duly considered while collecting materials for laboratory diagnosis.
2. All materials collected should be accompanied by a full history of disease outbreak namely species affected, duration of disease, clinical signs, morbidity and mortality rates, disease suspected etc.
3. The collected biological specimens should be transported on ice to the nearest laboratory as early as possible.
4. Materials collected for bacteriological examination should be kept at refrigeration temperature (4 °C) in case of delay of transportation. If a viral etiology is suspected the material can be stored at –20°C to -80 °C.
5. When sero-diagnosis is required, collect paired serum samples (about 2 ml sera). One serum sample should be collected at the onset of disease and second serum after recovery (3-4 weeks) from disease preferably on the 21st day.
6. If death is reported, the post-mortem examination should be conducted at the earliest as putrefied materials are unfit for laboratory examination & detailed post-mortem report should be attached along with the samples collected during post-mortem.
7. For Histopathological studies, tissues should be preserved in 10% formalin. The volume of formalin used should be approximately 10 times the volume of material. Specimen bottles with wide mouths should be used for collecting tissues.
8. All the impression smears before packing, should be fixed in methanol for 1-5 minutes unless otherwise specified.
9. In case of outbreaks, try to collect materials from as many ailing animals (5-6 or more) as possible at the height of body temperature /clinical signs.

**Notes on Disease and Disease control – Livestock Inspectors
Training**

Collection Procedure of different biological material /specimen

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Name of specimen

Collection Procedure

Blood

Blood: Blood is examined for different purposes as listed below :

- Preparation of blood smears for microscopic examination:
The ear vein is the most convenient site for taking blood when required for the preparation of smears or films. In poultry, blood is taken from comb, wattles or wing veins.
- For cultural examination, transmission experiments and other studies:

In cows, buffaloes, horses, sheep and goats, blood is preferably taken from the jugular vein.

From pig, blood is usually obtained by cutting the end of the tail or tip of the ear, or from anterior vena cava.

In dogs and cats, from the saphenous or radial vein and from the fowl's brachial vein (on the ventral surface of the wing). From ducks jugular vein is the preferred site.

The site should be prepared by removing hair /feathers and disinfecting it with spirit. Blood is collected by a sterile syringe. From post-mortem cases, blood is collected from the right ventricle.

For serological studies, serum is separated by allowing the blood to clot, keeping the syringe in a slanting position. After complete separation, serum is transferred to a sterile vial.

For haematological studies, blood is not allowed to clot by using anticoagulants (sodium citrate or EDTA). One part of anticoagulant is added to 9 parts of blood.

**Notes on Disease and Disease control – Livestock Inspectors
Training**

Preservation of materials for specific examination

Draft - L1 Notes

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Specific examination

Preservatives used

Bacteriological/ Mycological

- Blood and exudates: are to be collected aseptically with sterile Pasteur pipettes/syringes, and then put in sterile tubes or vials without any preservative and to be transported over ice. Especially for Anthrax and HS blood should be collected in sterile container /Cryo vials with (500-1000µl EDTA is sufficient) and without EDTA. After collection cotton soaked with alcohol should be placed on the collection site and burned and transported strictly under cold chain.
- Tissues: Pieces of affected organs with lesions should be collected in sterile condition and should be transported under cold chain.

Virological

- Blood and exudates: to be collected aseptically without any chemical preservative and to be stored and transported on ice. In some viral infections (e.g. Blue Tongue and Classical Swine Fever blood may be collected in vials/vacutainer containing EDTA).
- Tissues: Pieces of affected organs are to be collected under aseptic conditions and transported either on ice. For FMD 50% buffered glycerol saline is the preferred transport medium

Parasitological

- For identification of parasite and helminth ova 10% formalin is the preferred transport medium to preserve the integrity of the ova..
- For coccidial oocysts, 2.5% Potassium

Despatch of Samples to Animal Health Laboratories:

Following points must be kept in mind while despatching the material to laboratory for diagnosis:

1. Describe the clinical signs, lesions, tentative diagnosis and treatment given to animals in your letter; also mention the type of test you want with your tentative diagnosis.
2. Write correct address on letter as well as on the parcel preferably with pin code, if the material is sent through post.
3. Mark the parcel “Biological Material” “Handle with Care” “Glass Material” etc .In order to avoid damage in the parcel, also mark the side to be kept on the upper side with arrows.
4. Seal the container so that it should not leak in transit.
5. A separate specimen box should preferably be used for each case so that they do not contaminate each other, or leak in transit..
6. Try to send the material as soon as after collection from the animal.
7. Keep one copy of cover letter inside the parcel and another copy by hand or post in a separate cover
8. Put a frozen coolant pad into each tin when packing the specimens for despatch.
9. In poisoning cases the address of the Forensic Laboratory should be clearly written.

The bottle or container should be sealed and labelled properly indicating the name of owner, identification of animal (tag no /mark/colour/sex/ breed /age/species) type of tissue collected and preservative used (if any).The examination requested and disease or poisoning suspected should also be written.

- a. A copy with details of post-mortem report and containing above information should be sent separately under separate cover
- b. All the containers should be packed with cloth and sealed with sealing wax and should be sent through a person in order to avoid any breakage in transit.

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COMPOSITION OF COMMON PRESERVATIVES /TRANSPORT MEDIA

1. Normal Saline (pH 7.2) 8.5 gm of sodium chloride in 1 litre of distilled water.
2. 50% Glycerol Phosphate Buffer (pH 7.4): FMD Samples

Sodium chloride	8 gm
Potassium chloride	0.2 gm
Di sodium hydrogen phosphate	1.15 gm
Potassium dihydrogen phosphate	0.2 gm
Distilled water (make up to)	1000 ml
Glycerine (sterile)	1000 ml
3. 50 % Glycerol saline 500 ml

Normal saline	500 ml
Glycerine(sterile)	500 ml
4. 10% Buffered formalin

Formalin (commercially available 40% Formaldehyde)	100 ml
Sodium phosphate (monosodium monohydrate)	4 gm
Sodium phosphate (dibasic, anhydrous)	6.5 gm
Distilled water (make up to)	1000 ml
5. 10% Formalin/Formol saline

Formalin (commercially available 40% Formaldehyde)	100 ml
Distilled water/Normal saline	900 ml
6. Amies Transport Medium (Make up to 1000ml. Adjust pH to 7.2 & sterilize by autoclaving)

Sodium Chloride	3.0 gm
Potassium Chloride	0.2 gm
Calcium Chloride	0.1 gm
Magnesium Chloride	0.1 gm
Mono potassium phosphate	0.2 gm
Sodium thioglycolate	1.0 gm
Charcoal (optional)	10.0 gm

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Agar 4.0 gm

7. Fixatives for blood & tissue impression smears:

(2-3 dips or dipping for 2-3 minutes after air drying smear)

- Methanol (absolute)
- Ethyl alcohol (rectified spirit)/Isopropyl alcohol (absolute)

8. Transport medium for Molecular tests

- Ethyl alcohol (absolute)
- Commercially available transport medium (RNA later, Trizol, VTM)

9. Preservatives for Toxicological samples

- Saturated saline (NaCl)
- Rectified spirit

10. Microbiological Investigations

- Commercially available blood culture tubes, Viral Transport Medium (VTM)
- Amies Transport media(Himedia/Thermo Fisher/ Sigma)
- Nutrient broth/Peptone water/ Normal saline/PBS (sterile)

11. Parasitological Examination

10 % formalin for faecal samples

70 % alcohol for helminths, cysts, segments, ectoparasites

DISEASE CONTROL PROGRAMMES

OPERATIONAL GUIDELINES FOR NATIONAL ANIMAL DISEASE CONTROL PROGRAMME (NADCP) FOR FOOT AND MOUTH DISEASE (FMD) AND BRUCELLOSIS

Prevalence of animal diseases in India is a serious impediment to the growth of the Livestock Sector. Losses due to some of these diseases e.g. Foot & Mouth Disease (FMD), Brucellosis etc. are huge and often beyond estimation. FMD has a direct negative impact on trade of milk and other livestock products. Brucellosis is a reproductive disease of livestock resulting in huge financial losses and also having an adverse impact on human health. Farm workers and livestock

owners are always at risk of contracting as well as spreading this disease. It is therefore imperative to control FMD by vaccination of all susceptible species and Brucellosis by

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vaccination of all female bovine calves (4-8 months old) in the country. Advantage of this programme is healthy animals with better production and global acceptance of our animal products. Finally, efforts in this direction would further contribute towards doubling farmers income.

Control of FMD can be achieved by mass vaccination of susceptible livestock repeatedly at regular intervals till the incidence of the disease comes down. This will pave away to gradual control/eradication of the disease from the country.

Bovine brucellosis is endemic in India and appears to be on the increase in recent times, perhaps due to increased trade and rapid movement of livestock. The nature of livestock rearing, especially in rural India, is conducive to the spread of infection from livestock to humans due to close contact with animals. In the absence of any effective treatment in bovines, the disease can be prevented only by vaccination. Control of Brucellosis can be achieved by a once-in-a-lifetime vaccination of female bovine calves (4-8 months old).

Objectives of the programme

The overall aim of the National Animal Disease Control Programme (NADCP) for FMD and Brucellosis is to control FMD by 2025 by vaccination and its eventual eradication by 2030. Intensive Brucellosis Control programme in animals is envisaged for controlling Brucellosis which will result in effective management of the disease, in both animals and humans.

Programme implementation

NADCP for FMD and Brucellosis is a Central Sector Scheme where 100% of funds shall be provided by the Central Government to the State/Union Territories.

At the national level, overall implementation and monitoring of NADCP would be done by the following agencies as under-

Information Network for Animal Productivity and Health (INAPH) portal of National Dairy Development Board (NDDB). This portal shall serve as the central database for animals registered uniquely on INAPH. 100% central funding would be provided to NDDB for registering the animals (vaccinated) and maintaining the INAPH database.

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State/UT Level- The State/UT Government shall provide the requisite administrative support and the necessary infrastructure for cold-chain maintenance for vaccines, power back up etc. and manpower to carry out vaccination and other related activities in a systematic manner as per the planned calendar in order to effectively control these diseases. For smooth implementation of the programme at the district, block and village level, the State Animal Husbandry Department shall constitute the following units with their enlisted roles and responsibilities as under

States Monitoring Unit (SMU)

1. SMU shall ensure that vaccination programmes should be carried out in Mission mode in the shortest possible time period (within 21-30 days for FMD-CP) for its effectiveness so as to build herd immunity and ensure vaccination of all leftover animals and new introduction.
2. For FMD-CP, it must be ensured that all susceptible animals, including pregnant ones, must be included in the vaccination programme. Fear of abortion due to vaccination, if any, must be alleviated through extensive Information and Education Campaign at the farmer's level.
3. SMU shall also ensure complete vaccination of all stray animals to be covered under the programme.
4. SMU shall ensure availability of stock of emergency medicines for immediate use as anti-shock treatment in a suspected case of anaphylactic reaction, if any
5. For proper identification of vaccinated animals, SMU shall ensure that ear-tags, tagging machines, etc. are available before the start of vaccination. All vaccinated animals shall be ear-tagged and registered in the INAPH animal health module database.
6. SMU shall ensure training of technicians involved in the process of ear-tagging, entry of details required and uploading of the data in the INAPH server. Simultaneous issuing of animal health cards for recording details of vaccinated animals to animal owners may also be ensured. Vaccination cards need to be issued for individual animals, wherever the same has not been issued earlier, and handed over to the animal owner.
7. SMU shall regulate the entry of animals from other States only against a vaccination proof of 21 days before. Wider publicity should be given to encourage the farmers to get the newly

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introduced animals vaccinated, preferably before their entry to the State. If not, at least immediately on arrival.

District Monitoring Unit (DMU)

Block Monitoring Unit/Block Veterinary Officers: Following are the responsibilities of the Block Veterinary Officers

- a) They shall ensure the supply of vaccine to field officers as per the calendar of vaccination.
- b) Shall ensure availability of adequate vaccine stock as per eligible animal population and ensure maintenance of cold chain.
- c) Shall ensure all pre-requisite of the vaccination programme like trained manpower, vaccine carriers, syringes, needles, biological waste deposit bags, personal protective equipment (apron, hand glove, plain protecting glass), ear tags, ear tagging machines, transport of vaccine, etc, before starting actual vaccination in the block.
- d) Shall ensure that a separate disposable needle is used for the vaccination of each animal.
- d) The Officer in-charge of Veterinary Institutes with his team shall vaccinate all eligible animals as per programme norms.
- d) During the vaccination campaign, it should be ensured that vaccine bottles are continuously kept in vaccine carriers when not in immediate use so that cold- chain is maintained.
- e) Vaccinators shall ensure that vaccine bottles once opened(punctured) should be used on the same day itself to avoid deterioration of potency/quality of vaccine.
- f) Vaccinators shall take care to avoid spoilage of vaccines.
- g) If the animal becomes ferocious or gets disturbed while carrying out vaccination thereby leading to spilling of the vaccine, it shall be ensured that such animals are once again vaccinated properly.
- h) Vaccinator shall hand over the vaccination card, duly signed by the Veterinary Officer, to every animal owner after vaccinating the particular animal.

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- i) The Officer In-charge of Veterinary Institutes shall monitor the vaccination programme in the villages under its jurisdiction and shall ensure participation of LSGIs and local leaders and other key persons.
- j) He/she shall provide information about the control programme, its importance and impact to the villagers through discussion, leaflets, pamphlets, posters, visual aids during training and awareness programmes etc., so as to encourage the farmers to vaccinate their animals.
- k) Appropriate disposal of used/waste materials or any other bio-medical waste generated also need to be ensured.

Activities under NADCP for FMD and Brucellosis

Control of Foot and Mouth Disease:

- vaccinating the entire susceptible population of bovines, small ruminants (sheep and goats) and pigs at six-monthly intervals (mass vaccination against FMD)
- primary vaccination of bovine calves (4-5 months of age) and deworming one month prior to vaccination.
- serosurveillance/sero monitoring of animal population
- Parasitic control: Deworming before vaccination helps to get a better immune response. Deworming shall be done twice a year, preferably to commensurate with the vaccination programme, but 3-4 weeks prior to it.
- Establishment/Strengthening of check posts: FMD is a trans-boundary animal disease and hence, veterinary border check posts set up to control movement of live animals and animal products entering the country or inter-state is one of the ways to control animal movement.

Control of Brucellosis:

The major activities of this component include mass screening of cattle and buffaloes to know exact incidence of the disease in an area/villages/block/district of the state and once in lifetime vaccination of all female calves between 4-8 months using B. Abortus S-19 strain vaccine (any alternative vaccine may replace the existing one in future).

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Vaccine and vaccination: As Brucellosis in animals is highly zoonotic, handling of live attenuated vaccines and vaccinating animals need extra care. Any accidental exposure to it may result in infection to the personnel engaged in vaccination. The manpower physically engaged in vaccination of animals need personal protective equipment (PPE) gumboots etc.

Details of vaccination for FMD

Description of animal	Vaccination schedule
Young Animals	Bovine calves 4-5 months old. Booster dose of vaccine shall be administered four weeks after primary vaccination and regular vaccination should be followed thereafter every 6 months
Adult Animals	Six-monthly vaccination as recommended under the programme.
FMD vaccine shall be kept constantly at a temperature between 2° C and 8° C. The vaccine shall neither be frozen nor be exposed to temperature higher than 8° C. The dose of vaccine used shall be as per the manufacturer's instructions.	

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Details of work plan with technical indicators for NADCP- BCP		
Sl No	Items of Work plan	Technical indicator
1	States to be covered	All States and Union Territories in the Country
2	Animals to be vaccinated	100% vaccination coverage of bovine female calves of 4-8 months of age
3	Vaccination	Once in a life- time calf-hood vaccination
4	Vaccine	Brucella S-19
	Brucella vaccine shall be kept constantly at a temperature between 2° C and 8° C. Use only the diluent provided with the vaccine for its reconstitution. The dose of vaccine used shall be as per the manufacturer's instructions. which is at present 2ml each for female cattle and buffalo calves.	

SERO MONITORING

FMD pre and post vaccination sero monitoring under National Animal Disease Control Programme (NADCP)

Since many factors can influence the effectiveness of vaccination against FMD, the programmes must be monitored continuously to identify any failings and to ensure sustained control of the disease.

Monitoring of vaccination programmes and population immunity are important components of the surveillance system. They are also a requirement for those seeking official recognition by the World Organisation for Animal Health (OIE). Serological tests that detect antibodies are suited to measuring protective antibody responses induced by vaccination.

A bi-annual vaccination based FMD Control Programme (FMDCP), involves six monthly FMD vaccinations (with a trivalent FMD serotypes O, A and Asia 1 vaccine) of all cattle and buffaloes and estimation of level of serotype specific antibodies. Currently, this programme covers the

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entire country.

Sampling strategy followed for post vaccination monitoring

The whole process of sample collection (Pre-vaccination and post-vaccination sampling) is repeated again during each round of vaccination through a randomized calculator. However, it is important to note that, during each round of vaccination, the villages in a district have been changed to collect the pre and post vaccination serum samples. Therefore, this strategy has been providing a clear immunity figure of the whole state or country and the efficacy of both vaccine and vaccination programme.

Collection and transport of serum samples.

Serum should be collected using a sterile needle and syringe. At least 2 ml serum should be submitted to the testing laboratory and for that 4 - 8 ml whole blood needs to be collected without anticoagulant. The blood is usually collected from the jugular vein. After collection, the sample should be allowed to stand in a cool area, out of direct sunlight for at least 15 minutes for clot formation. Serum should be separated before submission to the laboratory. The sample should be centrifuged or, if this is not available, keep the sample in an upright position overnight in a cool box or refrigerator. Transfer serum into a plain, sterile tube and send it to the laboratory under cool condition. If dispatch to a laboratory is delayed, serum samples should be frozen and stored at 20° C. Samples should be accompanied by a proper epidemiological data sheet when submitted to the testing laboratory .

STATE ANIMAL DISEASE EMERGENCY RESPONSE CELL (SADEC)

SADEC functions as a round the clock working department under ADCP (HQ), AHD, Kerala. Here a Veterinary Surgeon, three LiveStock Inspectors and three Attendants are working in three shifts.

Major duties and responsibilities of SADEC : -

1. Animal Disease Surveillance (Panchayath wise Reports) : Collection and verification of information received from all the districts, preparation of consolidated monthly reports and forwarding monthly reports to the Director of Animal Husbandry and CDIO, SIAD, Palode.
2. Disease outbreaks: Collection and consolidation of daily reports received in the prescribed format from all the districts during outbreaks and forwarding to The Director of Animal Husbandry and Government of India
3. NADCP (FMDCP, BCP) and LSD vaccinations : Collection and consolidation of daily reports received from all the districts and forwarding to The DAH and GoI . Uploading daily reports in the DAHD dashboard during vaccination period.
4. African Swine Fever & Avian Influenza : Preparing outbreak reports in the prescribed format during outbreaks and forwarding daily reports to the Director of AH and Government of India.
5. Zoonotic diseases : Collection of data from all districts, preparation of consolidated daily reports and forwarding to the Director of AH and Media .
6. Round the clock contact point : Attending and managing phone calls (from farmers and AHD staff) 24/7.

